Scoped O.1 on the Prescribed Cone

Apex Heat Moments, a Constant-Profile Slavnov–Taylor Identity, and Domain–Readout Boundaries

Version 8; revision of `XVI-O1-CONE-v7.tex`:
executed replay, exact span certificates, and constant-profile maps

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Abstract

This is Version 8 of the scoped O.1 cone audit; it revises XVI-O1-CONE-v7. The cone $C_\alpha \times \mathbb{R}^2$, its deficit and holonomy, and the curvature-regular (CR) and componentwise Friedrichs (F) realizations remain prescribed inputs. The v7 analytical content is retained. What is new is executed and is stated with its scope.

(1) The v7 program was run. The embedded `cone_v7.py` was extracted unchanged and executed. Its self-test passed, all 27 replay runs returned 0, and the coverage-aware comparison with the historical logs passed for all 18 graph–branch pairs. Of the fifteen static corrections listed in v7, eight are real: seven code defects and one methodological limitation. None of them changes a number printed in v6.

(2) The count of 26 is now exact, and so is a complete span certificate. We evaluated the apex functionals on every relevant unit atom over a complete 6×6 momentum grid, in exact $\mathbb{Q}(\mathrm{i})$ arithmetic. At $\alpha = 3/10$ (CR and F) and at $\alpha = 7071/10000$ (CR), the inventory is $34 = 18 + 8 + 6 + 2$. The Slavnov–Taylor row rank is 26, with a nonzero minor and a kernel verified on all rows. The closed space is Z with $\dim Z = 8$. The exact directions B satisfy $\dim B = 6$ and $B \subset Z$, and $\dim H^0 = 2$. At $\alpha = 7071/10000$ the F branch is different: both the historical modular record and the v8 exact run give $38 = 21 + 8 + 7 + 2$, with ST rank 29, $\dim Z = 9$ and $\dim H^0 = 3$. The number 34 is therefore a branch- and angle-qualified count, not a universal one.

(3) Projections. Six of the 26 Ward combinations are supported on the twelve constant-profile coordinates. Two of them lie in the hh block: $b_1 + 2b_3 = 0$ and $b_2 = 0$. Both hold exactly on the hh catalogue sum. They hold through a cancellation between D01 and D03; neither graph satisfies them alone. The other four involve the CK/CB sectors, and we identify them with the v7 rows only conditionally. The remaining 20 are open. Of the 18 engine hh coordinates, the six constant-profile ones receive an explicit exact dictionary; the other twelve stay open. Of the six exact values, four combinations are fixed, including $e_1 = (1 - \alpha)/4$ and $e_3 = 0$. The CR/F difference sits only in $e_4 + e_5$, where it equals $\nu/8$ exactly at both computed angles. The remaining two combinations are open.

(4) New mixed-channel finding. The engine slice exposes hidden mixed pairs. These are pairs whose circle pole vanishes while their Feynman density does not. In D01, the lattice keys $(4,4), (4,7), (5,5), (7,4), (7,7)$ carry a mixed $(0,2)$ density proportional to $P^2(6x^2 - 10x + 3)$. The printed v6 hh branch difference is therefore unchanged in value. Its status becomes conditional on the corresponding mixed-channel trace. A per-mode Duhamel identity and a reduction formula are proved, but the D04 part of the second $\phi\phi$ value stays open.

(5) Evanescent coefficients. Under an explicitly declared slice-DRED prescription with formal $\tilde{g}$ components, the number of evanescent pole coefficients is 86 on each physical branch and 113 on the UNIT lattice. They are prescription conditional.

No missing value is set to zero. The exact rank statement has status `CLOSED_PENDING_INDEPENDENT_REVIEW` only on its precisely stated finite scope. Overall:

$$\text{O.1 on the cone} = \texttt{NOT_CLOSED}, \qquad \texttt{bulk_symmetry_restored} = \texttt{false}.$$

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1 Question, evidence, and revision contract

1.1 The scoped question

The target is transport of the separately scoped XVI-Q two-point record to a prescribed conical product. The fixed scope is:

(i) one loop and coupling order $O(\kappa^2)$;

(ii) the two-point sector with vanishing declared scalar mass;

(iii) frozen de Donder gauge;

(iv) dimensional reduction, with quasi-d loop momenta and quasi-four-dimensional vertex indices;

(v) the XVI-P p22 contract and the declared graph conventions;

(vi) a comparison of CR and F realizations without selecting either by a finite readout.

The later p23 excerpt is not silently substituted for p22. The frozen XVI-P, XVI-Q, and XVI-R records are source-qualified inputs and comparisons [4, 5, 6]. The immediate sources are the supplied v6 manuscript and its script collection [2, 3].

The central analytical question is not merely whether a conical heat coefficient exists. It is whether a complete differentiated graph, on its specified operator domain and in its specified meromorphic prescription, reduces to that coefficient and satisfies the required interacting identities.

1.2 Evidence classes

Analytical proof.
A statement proved below from the displayed definitions and explicit mathematical inputs. Its proof does not depend on a claim that a program was run.

Declared catalogue.
The explicit arrays in Section 8. Algebraic identities of these arrays are proved locally. Identification with complete cone graph residues is conditional on Assumption 7.1.

Historical executed record.
The v6 source reports executions, archived logs, and SHA-256 comparisons. Those reports are preserved as attribution to v6, not represented as a new execution in this revision.

Embedded verification program.
Appendix A contains a consolidated, corrected program whose purpose and acceptance conditions are stated explicitly. It is supplied as source, not as an executed certificate. In v8 it was extracted unchanged and executed (Section 13.2).

Executed in v8.
A statement whose evidence is a v8 run on the I-B audit box: a named script under `code/v8/`, its return code, and a log under `logs/v8/`, all hashed in Appendix B. Exact runs use $\mathbb{Q}(\mathrm{i})$ arithmetic; numerical runs are labelled corroboration. An executed statement about the declared catalogue remains conditional on Assumption 7.1.

Documentary comparison.
A version-qualified statement from another manuscript, used to keep models and targets distinct. Such a citation is not an additional graph hypothesis unless explicitly declared as one.

What is self-contained here

The heat-moment proofs, the closed Mellin calculation, the catalogue identities, the finite-channel separations, and the conditional graph-reduction theorem are self-contained at the level stated. The frozen vertex arrays and raw historical logs are not embedded in the material supplied for this revision. Consequently this single LaTeX file is not a self-contained rebuild of the complete loop calculation. The source listings make routing and reduction logic inspectable; they do not manufacture the missing input arrays.

1.3 What v7 changes

The revision makes four substantive improvements.

(a) It places the audit on the I-B host and gives a standing domain ledger.

(b) It proves the D01 Mellin value in closed form, without requiring a symbolic simplifier or numerical quadrature.

(c) It corrects several overstrong or ambiguous statements: reflection-even invariance, uniqueness of extension, mixed-key coverage, modular certificates, and numerical evidence.

(d) It promotes the continuum/preparation/instrument split and the I-C A–F separations into the release boundary.

The original coefficient values are retained. No new complete cone graph value is claimed.

1.4 What v8 changes

Version 8 keeps the v7 structure and every v7 proof. It adds executed evidence, and every executed statement names its artifact (Appendix B).

(a) The v7 listing was extracted and run. Each of the fifteen v7 corrections receives a verdict (Section 13.2).

(b) An exact, complete span certificate replaces the earlier modular and random-rank evidence. It fixes the count of 26 Ward combinations and the inventory dimension, each at named (α, branch) (Section 14).

(c) The 34-candidate basis is projected onto its twelve constant-profile coordinates. The hh part of the 18-dimensional engine map is made explicit on six of its 18 coordinates, and four of the six exact values are fixed (Section 15).

(d) Hidden mixed-channel pairs are found. As a result the v6 hh branch difference becomes conditional. A Duhamel lemma and a reduction formula for mixed traces are proved (Section 16).

(e) Evanescent pole coefficients are computed under a declared slice-DRED prescription (Section 17).

(f) The open-obligation table and the release theorem are re-stated with the labels OPEN, PARTIAL, CONDITIONAL and `CLOSED_PENDING_INDEPENDENT_REVIEW`.

No v6 number changes value. One v6 statement changes status, from unconditional on the declared catalogue to conditional, as described in (d).

1.5 Programme non-transfer

The cone and CR branch remain inputs. No Einstein dynamics, physical tensor Hessian, spacetime emergence, gravitational Page curve, or inherited quantum coefficient is constructed. XVI-N is

not discharged; the downstream XV-B admission exit is not opened. The historical H2 ending is retained version-specifically, the narrower I2 construction is not withdrawn, and the full I3/J2 characteristic maps remain separate [25, 26, 27].

The extended Hawk05 discussion supplied with the revision request concerns finite access and Page proxies. It is relevant only to the reporting boundary below; it is not imported as an apex calculation or as gravitational dynamics [24].

2 The prescribed I-B host and the typed domain ledger

2.1 Geometry

The Euclidean spectral host is

$$X_\alpha = C_\alpha \times \mathbb{R}^2, \qquad \mathrm{d}s_E^2 = \mathrm{d}r^2 + \alpha^2 r^2 \mathrm{d}\varphi^2 + \mathrm{d}y_0^2 + \mathrm{d}y_3^2, \quad 0 \le \varphi < 2\pi, \quad \alpha > 0. \tag{2.1}$$

The transverse measure is

$$\mathrm{d}V_\perp = \alpha r \, \mathrm{d}r \, \mathrm{d}\varphi.$$

The apex stratum in the completed product is

$$\mathcal{W} = \{r = 0\} \times \mathbb{R}^2.$$

For the deficit comparison we use $0 < \alpha \le 1$, with deficit $2\pi(1-\alpha)$.

The Lorentzian regular host has $r > 0$ and metric

$$\mathrm{d}s_L^2 = \mathrm{d}t^2 - \mathrm{d}r^2 - \alpha^2 r^2 \mathrm{d}\varphi^2 - \mathrm{d}z^2. \tag{2.2}$$

This is the I-B prescribed conical host, with an overall signature convention chosen to agree with the pole catalogue [7]. The causal geometry is unchanged by the opposite overall signature used in some companion descriptions.

Proposition 2.1 (The regular host is not globally hyperbolic)**.** *The spacetime* (2.2) *with its apex removed is not globally hyperbolic.*

Proof. Fix $r_* > 0$, an angle, and a longitudinal position. Choose $a > r_*$, and let p and q be the corresponding points at times $-a$ and a. For any sequence $r_j \downarrow 0$, the points at time zero and radius r_j, with the same angle and longitudinal position, lie in $J^+(p) \cap J^-(q)$: the radial spatial distance is less than a. They have no convergent subsequence in the regular host. The causal diamond is therefore not compact. □

This is the same geometric obstruction recorded in I-B-G [8, `prop:nonGH`]. A free inverse is accordingly tied to an extension and a test domain. Ordinary globally hyperbolic uniqueness must not be substituted for those data.

The apex is a worldsheet defect, not a derived horizon. There is no state-dependent metric update.

2.2 Standing host and domain ledger

Table 1: Standing comparison ledger. Companion claims are used only within the indicated source scope.

Input or comparison	Source	Permitted use	Excluded inference
Prescribed cone, deficit, and holonomy	I-B [7]	Host geometry and installed parameters	A horizon, metric update, or matter-driven sourcing law.
Scalar Friedrichs Green operator and finite speed	I-B-G, `thm:green`, `thm:speed` [8]	Free inverse and support control on its declared domain	An exact sequence on ordinary compact-spacetime tests; `rem:compact-exact` explicitly retains that gap.
Maxwell–Friedrichs photon with scalar ghost	I-B-G-S, `thm:spin1` [9]	Source-tested free reference	An interacting Slavnov–Taylor identity or discharge of a named cone Ward combination.
CR source-test selection	I-B-G-S, `def:geo`, `thm:selection`, `thm:spin2` [9]	Declared mathematical comparison branch	A physical smooth-core selection theorem.
Positive tensor-core exclusion of idealized CR	I-B-G-S-B, `cor:exclude`, `thm:contrast`, `prop:warddefect` [10]	Source-qualified comparison for its declared tensor families	An independently completed tensor proof in this paper, or a choice made by the present ST row.
Friedrichs Wick square, stress, and holonomy-odd current	I-B-VP and I-B-VP-1 [11, 12]	One-point and compact-test infrared comparison	Absence of infrared poles in massless loop graphs, or a two-point graph prescription.
Frozen-source Maxwell response	I-B-VP-4 [13]	Classical Maxwell comparison, if such a detector is declared	An interacting Ward identity.
Smooth-core finite parts and scalar/Dirac selection	I-B-VP-5-Ex1-B, Ex2, Ex2-B [14, 15, 16]	Qualifications of core limits and finite-part universality	An interacting spin-2 CR graph calculation.

The CR/F row proved below is labelled against this table: it is a constant-profile catalogue identity on two prescribed branches. It is neither the free photon theorem nor a tensor-core selection theorem.

Remark 2.2 (Sourcing is a separate equation). I-B's `prop:not-source` separates symmetry, conservation, and trace conditions from an Einstein sourcing equation [7]. For example, on a flat chart a nonzero constant symmetric conserved traceless tensor need not equal the Einstein tensor of that chart. The same refusal applies to every heat coefficient below.

2.3 Self-adjointness is not destructive access

For a radial Bessel expression

$$L_\nu = -\partial_r^2 - r^{-1}\partial_r + \nu^2 r^{-2}$$

in $L^2((0,\infty), r\,\mathrm{d}r)$, the endpoint pairing is

$$\mathfrak{b}(u,v) = \lim_{r\downarrow 0} r\big(\overline{u}\,v' - \overline{u'}\,v\big). \tag{2.3}$$

Proposition 2.3 (Closed endpoint dynamics)**.** *A self-adjoint realization A generates a unitary Schrödinger group $e^{-\mathrm{i}tA}$. On sufficiently regular domain solutions its total endpoint probability flux vanishes. A bound eigenstate is not an absorbing output channel.*

Proof. The spectral theorem gives unitarity. Equivalently,

$$\frac{\mathrm{d}}{\mathrm{d}t}\|u(t)\|^2 = 2\,\mathrm{Re}\langle u(t), -\mathrm{i}Au(t)\rangle = 0.$$

The endpoint current is proportional to $\mathrm{Im}(r\overline{u}\,u')$, which vanishes in the total boundary pairing on a self-adjoint domain. An eigenstate evolves by a phase. □

These are standard endpoint facts [32, 33]. They do not identify ghosts or gauge-fixed tensor components with a positive physical probability space.

Self-adjointness alone does not imply positive-energy wave stability: a negative eigenvalue $-\sigma^2$ produces $\ddot{a} - \sigma^2 a = 0$. Conversely, reduced access can be lossy while an enlarged complete dynamics remains unitary. Neither an absorbing boundary nor an apex reservoir is introduced here.

3 Twisted scalar channels and the normalized apex trace

3.1 Channels and subtraction

For $b \in [0, 1)$, the twisted Friedrichs transverse Laplacian A_b^{F} has radial orders

$$\nu_n = \frac{|n+b|}{\alpha}, \qquad n \in \mathbb{Z}.$$

It chooses r^{ν_n} at zero, with the no-logarithm condition when $\nu_n = 0$.

Its diagonal heat kernel, relative to $\mathrm{d}V_\perp$, is

$$K_b^{\mathrm{F}}(s; r, \varphi; r, \varphi) = \frac{e^{-z}}{4\pi\alpha s} \sum_{n\in\mathbb{Z}} I_{|n+b|/\alpha}(z), \qquad z = \frac{r^2}{2s}. \tag{3.1}$$

We use the Bessel heat representation and standard modified Bessel identities [31, 32].

The infinite cone has infinite ordinary heat trace. The apex functional is therefore defined by subtraction, not by writing an unqualified L^2-trace.

Definition 3.1 (Abel-subtracted apex functional)**.** For $u, s > 0$, let

$$T_b(u) = \int_{C_\alpha} e^{-(\cosh u - 1)r^2/(2s)} K_b^{\mathrm{F}}(s; x, x)\,\mathrm{d}V_\perp, \tag{3.2}$$

$$T_{\mathrm{bulk}}(u) = \frac{\alpha}{2(\cosh u - 1)}. \tag{3.3}$$

Define

$$\mathrm{Tr}_{\mathrm{ap}}\, e^{-sA_b^{\mathrm{F}}} = \lim_{u\downarrow 0}(T_b(u) - T_{\mathrm{bulk}}(u))$$

when the limit exists.

Theorem 3.2 (Normalized twisted apex constant)**.** *For every $\alpha > 0$, $b \in [0, 1)$, and $s > 0$,*

$$\mathrm{Tr}_{\mathrm{ap}}\, e^{-sA_b^{\mathrm{F}}} = \mathcal{C}_b^{\mathrm{F}} = \frac{1}{12}\left(\frac{1}{\alpha} - \alpha\right) - \frac{b(1-b)}{2\alpha} = \frac{B_2(b)}{2\alpha} - \frac{\alpha}{12}, \tag{3.4}$$

where $B_2(b) = b^2 - b + \frac{1}{6}$.

For a compact proper-time superposition

$$f(\lambda) = \int_{s_-}^{s_+} w(s) e^{-s\lambda}\,\mathrm{d}s, \qquad 0 < s_- < s_+ < \infty, \quad w \in L^1,$$

the corresponding apex functional satisfies

$$\mathrm{Tr}_{\mathrm{ap}}\, f(A_b^{\mathrm{F}}) = \mathcal{C}_b^{\mathrm{F}} f(0). \tag{3.5}$$

Proof. For $\rho > -1$ and $u > 0$,

$$\int_0^\infty e^{-z\cosh u} I_\rho(z)\,\mathrm{d}z = \frac{e^{-\rho u}}{\sinh u}.$$

The nonnegative Bessel summands permit interchange with the radial integral. Summing the geometric series gives

$$T_b(u) = \frac{\cosh((\frac{1}{2}-b)u/\alpha)}{2\sinh u\,\sinh(u/(2\alpha))}. \tag{3.6}$$

Its Laurent expansion is

$$T_b(u) = \frac{\alpha}{u^2} + \frac{B_2(b)}{2\alpha} - \frac{\alpha}{6} + O(u^2),$$

whereas

$$T_{\mathrm{bulk}}(u) = \frac{\alpha}{u^2} - \frac{\alpha}{12} + O(u^2).$$

Subtraction proves (3.4). The change of radial variable removed s, so the result is independent of proper time.

On the compact proper-time interval the subtracted expressions are uniformly bounded for sufficiently small u. Dominated convergence and linearity give

$$\mathrm{Tr}_{\mathrm{ap}}\, f(A_b^{\mathrm{F}}) = \mathcal{C}_b^{\mathrm{F}} \int_{s_-}^{s_+} w(s)\,\mathrm{d}s.$$

This is (3.5). □

Host boundary. The background and realization are prescribed. A heat coefficient, conservation identity, or trace identity does not imply that the corresponding object sources the cone.

The untwisted result and the conical heat-kernel method are classical [28, 29, 30]. The calculation above fixes the normalization actually used here. It is not an unrestricted ordinary-trace theorem for arbitrary resolvent products or differentiated graph kernels.

3.2 An image contribution is not a diffracted-front report

Example 3.3 (A pure-image apex contribution)**.** At $\alpha = \frac{1}{2}$, $b = 0$,

$$\mathcal{C}_0^{\mathrm{F}} = \frac{1}{8}.$$

The nonidentity image in the two-fold plane quotient contributes

$$\frac{1}{4\pi s} e^{-r^2/s}.$$

Integrating over its wedge of angle π gives

$$\frac{\pi}{4\pi s}\int_0^\infty r e^{-r^2/s}\,\mathrm{d}r = \frac{1}{8}.$$

Thus an apex heat constant can be nonzero in an image representation without requiring a separately nonzero strictly diffracted-front contribution.

The image-isolated windows and short-window deficit sensitivity of I-B-ECHO concern another functional [17]. They do not identify a window ratio with $\mathcal{C}_b^{\mathrm{F}}$ or with a graph residue. This separation is part of the release theorem, not merely an informal analogy.

4 Exceptional Bessel realizations and CR/F coefficients

4.1 A one-channel replacement

For $0 < \nu < 1$, let A_ν^+ and A_ν^- be the nonnegative homogeneous regular and singular Bessel realizations in $L^2(r\,\mathrm{d}r)$. Their radial heat kernels are

$$k_\nu^\pm(s; r, r') = \frac{1}{2s} e^{-(r^2+r'^2)/(4s)} I_{\pm\nu}\left(\frac{rr'}{2s}\right).$$

The endpoint choices, rather than a periodic relabelling of an ordinary fractional holonomy, distinguish them [32].

Theorem 4.1 (Relative diagonal heat shift)**.** *For $0 < \nu < 1$,*

$$\int_0^\infty (k_\nu^-(s; r, r) - k_\nu^+(s; r, r)) r\,\mathrm{d}r = \nu. \tag{4.1}$$

The relative diagonal integral is absolutely convergent.

Proof. It equals

$$\frac{1}{2}\int_0^\infty e^{-z}\big(I_{-\nu}(z) - I_\nu(z)\big)\,\mathrm{d}z.$$

At zero the worst behavior is $z^{-\nu}$, which is integrable. At infinity use

$$I_{-\nu}(z) - I_\nu(z) = \frac{2\sin(\pi\nu)}{\pi} K_\nu(z).$$

The resulting integrand decays exponentially. With the Abel factor, the integral is

$$\frac{e^{\nu u} - e^{-\nu u}}{2\sinh u} = \frac{\sinh(\nu u)}{\sinh u}.$$

Dominated convergence gives ν. □

Host boundary. The background and realization are prescribed. A heat coefficient, conservation identity, or trace identity does not imply that the corresponding object sources the cone.

4.2 Geometric helicities

For geometric helicity S, the Friedrichs fractional label is

$$b_{\mathrm{F}}(S) = \{S\alpha\}.$$

In the comparison chamber $1/2 < \alpha < 1$, put

$$\nu = 2 - \alpha^{-1} \in (0, 1). \tag{4.2}$$

For each exceptional helicity $S = \pm 2$, CR replaces one regular channel of order ν by the singular realization.

The per-helicity coefficients are

$$c_0 = \frac{1-\alpha^2}{12\alpha}, \qquad c_1 = \frac{5\alpha^2 - 6\alpha + 1}{12\alpha},$$
$$c_2^{\mathrm{F}} = \frac{23\alpha^2 - 36\alpha + 13}{12\alpha}, \qquad c_2^{\mathrm{CR}} = \frac{23\alpha^2 - 12\alpha + 1}{12\alpha}. \tag{4.3}$$

They satisfy

$$c_1 - c_0 = \frac{\alpha - 1}{2}, \qquad c_2^{\mathrm{CR}} - c_0 = 2\alpha - 1,$$

$$c_2^{\mathrm{F}} - c_0 = \frac{(\alpha-1)(2\alpha-1)}{\alpha}, \qquad c_2^{\mathrm{CR}} - c_2^{\mathrm{F}} = \nu. \tag{4.4}$$

These follow from Theorem 3.2 at $b = 0, \alpha, 2\alpha - 1$, and from Theorem 4.1. The polynomial substitution $b = 2\alpha$ reproduces c_2^{CR}, but is justified here only through that specified channel replacement.

At $\alpha = \frac{1}{2}$, continuous evaluation gives

$$c_0 = c_2^{\mathrm{F}} = c_2^{\mathrm{CR}} = \frac{1}{8}, \qquad c_1 = -\frac{1}{8}.$$

For $0 < \alpha < \frac{1}{2}$, the Friedrichs weight-two label is 2α, not $2\alpha - 1$; the displayed F polynomial must not be extended unchanged across that chamber boundary.

Remark 4.2 (The CR flat limit is not an endpoint realization). As $\alpha \uparrow 1$,

$$c_0, c_1, c_2^{\mathrm{F}} \longrightarrow 0, \qquad c_2^{\mathrm{CR}} \longrightarrow 1.$$

This is a one-sided polynomial limit. At $\nu = 1$, $r^{-1} \notin L^2(r\,\mathrm{d}r)$ near zero, so the deficient singular branch used in the proof is no longer available in the same form. No CR operator value at the exactly smooth endpoint is inferred. Spin-dependent nonuniform smooth limits also occur in the established conical heat-kernel literature [29].

5 The weighted apex moment

Definition 5.1 (Weighted Abel subtraction). For $u, s > 0$, define

$$T_b^{(2)}(u) = \int_{C_\alpha} r^2 e^{-(\cosh u - 1)r^2/(2s)} K_b^{\mathrm{F}}(s; x, x)\,\mathrm{d}V_\perp,$$
$$T_{\mathrm{bulk}}^{(2)}(u) = \frac{\alpha s}{(\cosh u - 1)^2}.$$

Their subtracted limit defines $\mathrm{Tr}_{\mathrm{ap}}[r^2 e^{-sA_b^{\mathrm{F}}}]$.

Theorem 5.2 (Normalized r^2-weighted coefficient). *For $\alpha > 0$, $b \in [0, 1)$, and $s > 0$,*

$$\mathrm{Tr}_{\mathrm{ap}}[r^2 e^{-sA_b^{\mathrm{F}}}] = s\,\mathcal{D}_b, \qquad \mathcal{D}_b = -\frac{11\alpha}{180} + \frac{B_2(b)}{3\alpha} - \frac{B_4(b)}{6\alpha^3}, \tag{5.1}$$

where $B_4(b) = b^4 - 2b^3 + b^2 - \frac{1}{30}$. Equivalently,

$$\mathcal{D}_b = \frac{1 + 10\alpha^2 - 11\alpha^4 - 60\alpha^2 b(1-b) - 30b^2(1-b)^2}{180\alpha^3}. \tag{5.2}$$

For the proper-time superpositions of Theorem 3.2,

$$\mathrm{Tr}_{\mathrm{ap}}[r^2 f(A_b^{\mathrm{F}})] = -\mathcal{D}_b f'(0). \tag{5.3}$$

Proof. The radial variable $z = r^2/(2s)$ gives

$$\frac{T_b^{(2)}(u)}{s} = -\frac{2}{\sinh u}\frac{\mathrm{d}}{\mathrm{d}u} T_b(u).$$

To expose the constant term, write

$$T_b(u) = \frac{\alpha}{u^2} + t_0 + t_2 u^2 + O(u^4),$$

where expansion of (3.6) yields

$$t_2 = \frac{B_4(b)}{24\alpha^3} - \frac{B_2(b)}{12\alpha} + \frac{7\alpha}{360}.$$

Therefore

$$\frac{T_b^{(2)}(u)}{s} = \frac{4\alpha}{u^4} - \frac{2\alpha}{3u^2} + \frac{7\alpha}{90} - 4t_2 + O(u^2).$$

The bulk expression is

$$\frac{\alpha}{(\cosh u - 1)^2} = \frac{4\alpha}{u^4} - \frac{2\alpha}{3u^2} + \frac{11\alpha}{180} + O(u^2).$$

The pole terms cancel and the constant difference is (5.1). Expanding the Bernoulli polynomials gives (5.2).

Dominated convergence on the compact proper-time support gives

$$\mathrm{Tr}_{\mathrm{ap}}[r^2 f(A_b^{\mathrm{F}})] = \mathcal{D}_b \int s\, w(s)\, \mathrm{d}s = -\mathcal{D}_b f'(0).$$

□

Host boundary. The background and realization are prescribed. A heat coefficient, conservation identity, or trace identity does not imply that the corresponding object sources the cone.

Proposition 5.3 (Weighted singular-channel shift)**.** *Replacing one regular channel of order $0 < \nu < 1$ by its singular realization changes the weighted coefficient by*

$$\Delta\mathcal{D} = \frac{2\nu(1-\nu^2)}{3}. \tag{5.4}$$

In the weight-two comparison chamber,

$$\mathcal{D}^{\mathrm{F}}_{S=2} = \mathcal{D}_{2\alpha-1}, \qquad \mathcal{D}^{\mathrm{CR}}_{S=2} = \mathcal{D}_{2\alpha-1} + \Delta\mathcal{D} = \mathcal{D}_{2\alpha}.$$

The last expression is polynomial evaluation, not an assertion that 2α is a fractional twist.

Proof. For one channel the weighted relative Abel integral divided by s is

$$-\frac{1}{\sinh u}\partial_u\left(\frac{2\sinh(\nu u)}{\sinh u}\right).$$

Its limit is (5.4). The weighted diagonal relative integral is convergent at zero because its Bessel integrand is $O(z^{1-\nu})$, and it is exponentially convergent at infinity. Substitution in (5.2) proves the final equality. □

In particular,

$$\mathcal{D}_0 = \frac{(1-\alpha^2)(1+11\alpha^2)}{180\alpha^3}, \quad \mathcal{D}_{S=1} = \mathcal{D}_\alpha, \quad \mathcal{D}^{\mathrm{F}}_{S=2} = \mathcal{D}_{2\alpha-1}, \quad \mathcal{D}^{\mathrm{CR}}_{S=2} = \mathcal{D}_{2\alpha}.$$

All four tend to zero as $\alpha \uparrow 1$. Thus the weighted CR polynomial has no flat remnant. This vanishing does not itself establish convergence of the singular operator family at the endpoint.

Theorem 5.4 (Exact orbifold and twisted-orbifold identities)**.** *For every integer $N \geq 2$ and $j \in \{0, \ldots, N-1\}$,*

$$\mathcal{D}_{j/N}(1/N) = \frac{1}{4N}\sum_{k=1}^{N-1} \frac{\cos(2\pi jk/N)}{\sin^4(\pi k/N)}. \tag{5.5}$$

Consequently,

$$\mathcal{D}_{j/N}(1/N) = \frac{(N^2-1)(N^2+11) - 60j(N-j) - 30j^2(N-j)^2}{180N}. \tag{5.6}$$

Proof. On the rotation quotient, the k-th nonidentity image has diagonal kernel

$$(4\pi s)^{-1}e^{-r^2\sin^2(\pi k/N)/s}.$$

The character contributes the corresponding holonomy phase. Pairing conjugate images replaces the phase by its cosine. The wedge has angle $2\pi/N$, and direct integration gives

$$\frac{2\pi/N}{4\pi s}\int_0^\infty r^3 e^{-r^2\sin^2(\pi k/N)/s}\,\mathrm{d}r = \frac{s}{4N\sin^4(\pi k/N)}.$$

The identity image is precisely the bulk subtraction. The character sector has orders $|Nn+j|$, hence $\alpha = 1/N$, $b = j/N$. Theorem 5.2 proves (5.5). Substitution in (5.2) proves (5.6). □

Examples include

$$\mathcal{D}_0(1/2) = \frac{1}{8}, \qquad \mathcal{D}_{1/3}(1/3) = -\frac{4}{27}, \qquad \mathcal{D}_{1/4}(1/4) = -\frac{1}{16}, \qquad \mathcal{D}_{2/5}(1/5) = -\frac{16}{25}.$$

Remark 5.5 (Corrected literature normalization). Fursaev uses the full opening angle $\theta = 2\pi\alpha$. His coefficient $C_4^{\mathrm{Fur}}(\theta)$, in equation (2.15) of [28], is

$$C_4^{\mathrm{Fur}}(2\pi\alpha) = \frac{(\alpha^{-2}-1)(\alpha^{-2}+11)}{90}.$$

The normalization relation is

$$\boxed{\mathcal{D}_0 = \frac{\alpha}{2}C_4^{\mathrm{Fur}}(2\pi\alpha).}$$

This replaces the memory-qualified identification in v6. It follows also by inserting $f = r^2$ in that reference's weighted heat expansion. No numerical spin-two coefficient is imported from the literature.

6 The D01 derivative obstruction and its Mellin value

6.1 The actual obstruction

Let ν be (4.2). The exceptional relative radial kernel is

$$\Delta k_s(r,r') = \frac{\sin(\pi\nu)}{\pi s}e^{-(r^2+r'^2)/(4s)}K_\nu\left(\frac{rr'}{2s}\right).$$

Consider the raising expression

$$X = \partial_r - \frac{\nu}{r}, \qquad \mathcal{R}_s(r) = \left[X_r X_{r'}\Delta k_s(r,r')\right]_{r'=r}.$$

Theorem 6.1 (Local obstruction)**.** *For every $0<\nu<1$,*

$$\mathcal{R}_s(r) = a_0\, r^{-2\nu-2} + O(r^{-2\nu}), \qquad a_0 = \frac{2\nu^2(4s)^\nu}{s\,\Gamma(1-\nu)} > 0. \tag{6.1}$$

Thus the ordinary diagonal integral $\int_0^\epsilon r\mathcal{R}_s(r)\,\mathrm{d}r$ diverges. The expression X does not map the full singular Bessel domain into the next L^2 channel.

Proof. The leading singular part of Δk_s is

$$\frac{(4s)^\nu}{2s\,\Gamma(1-\nu)}(rr')^{-\nu}.$$

Applying X_r and $X_{r'}$ multiplies it by $4\nu^2/(rr')$, proving (6.1). Moreover

$$X(r^{-\nu}) = -2\nu r^{-\nu-1}, \qquad \int_0^\epsilon r|r^{-\nu-1}|^2\,\mathrm{d}r = \infty.$$

This excludes the unrestricted domain-preserving intertwining statement. It does not exclude all densely restricted operators or every possible distributional extension. □

6.2 A closed Mellin formula

Theorem 6.2 (Exact Mellin continuation)**.** *For* $\operatorname{Re} z > 2\nu$*, define*

$$M_s(z) = \int_0^\infty r^{1+z}\mathcal{R}_s(r)\,\mathrm{d}r.$$

With $t = z/2$*, its meromorphic continuation is*

$$\boxed{M_s(z) = \frac{\sin(\pi\nu)}{2\sqrt{\pi}}\, s^{t-1}\, \frac{t\,\Gamma(t+\nu+2)\Gamma(t-\nu)}{\Gamma(t+\frac{3}{2})}.} \tag{6.2}$$

It is regular at $z = 0$ *for* $0 < \nu < 1$*, and*

$$M_s(0) = 0, \qquad M_s'(0) = -\frac{\nu+1}{2s}. \tag{6.3}$$

Possible poles are contained in

$$\{2\nu - 2j : j \in \mathbb{N}_0\} \;\cup\; \{-2\nu - 4 - 2j : j \in \mathbb{N}_0\}.$$

Some listed poles can cancel.

Proof. Put $x = r^2/(2s)$ and $A = \sin(\pi\nu)/(\pi s)$. Direct differentiation gives

$$\mathcal{R}_s(r) = \frac{A}{2s}e^{-x}H_\nu(x),$$

where

$$H_\nu = xK - 2xK' + K' + xK'' - 2\nu(K' - K) + \frac{\nu^2}{x}K, \qquad K = K_\nu(x).$$

The Bessel equation $xK'' + K' = (x + \nu^2/x)K$ reduces this to

$$H_\nu = 2\left[(x+\nu)K - (x+\nu)K' + \frac{\nu^2}{x}K\right].$$

For $\operatorname{Re}\mu > |\operatorname{Re}\nu|$, let

$$I_\nu(\mu) = \int_0^\infty x^{\mu-1}e^{-x}K_\nu(x)\,\mathrm{d}x = \frac{\sqrt{\pi}\,\Gamma(\mu+\nu)\Gamma(\mu-\nu)}{2^\mu\Gamma(\mu+\frac{1}{2})}. \tag{6.4}$$

For completeness, use

$$K_\nu(x) = \frac{1}{2}\int_0^\infty y^{\nu-1}e^{-x(y+y^{-1})/2}\,\mathrm{d}y.$$

Integrating in x first gives

$$I_\nu(\mu) = 2^{\mu-1}\Gamma(\mu)\int_0^\infty \frac{y^{\mu+\nu-1}}{(1+y)^{2\mu}}\,\mathrm{d}y.$$

The beta integral and the gamma duplication formula give (6.4). This also identifies the $a = 1$ limit of the standard Laplace–Bessel formula in DLMF 10.43.22 [31].

In the initial convergence half-plane, integration by parts has no endpoint contribution. It yields

$$M_s(z) = A(2s)^t\big[(t+1)I_\nu(t+1) + \nu(\nu+t)I_\nu(t)\big].$$

Since

$$\frac{I_\nu(t+1)}{I_\nu(t)} = \frac{t^2-\nu^2}{2t+1},$$

the bracket is

$$\frac{t(t+\nu)(t+\nu+1)}{2t+1} I_\nu(t).$$

Substitution of (6.4) proves (6.2) on a nonempty convergence half-plane and hence by meromorphic continuation.

At $t = 0$, the gamma factors are finite for $0 < \nu < 1$, and the explicit factor t gives $M_s(0) = 0$. Using

$$\Gamma(\nu+2)\Gamma(-\nu) = -\frac{\pi(\nu+1)}{\sin(\pi\nu)}$$

gives the derivative in (6.3). The pole statement follows from the numerator gamma functions. □

6.3 What is and is not unique

Remark 6.3 (Correction of the extension claim). The *Mellin prescription* is unambiguous once its initial integral and meromorphic continuation are specified. On smooth rotationally invariant transverse tests, the local Taylor powers are even. The singular heads $r^{-2\nu-2+2j}$ therefore have no resonance at $z = 0$ for $0 < \nu < 1$. In particular, $\nu = \frac{1}{2}$ is not a resonance of this even radial test class.

This does *not* prove that the full heat-time-dependent diagonal has a unique scale-covariant extension. At fixed s, $\mathcal{R}_s$ is not a homogeneous function of r. If δ_{rad} denotes the endpoint evaluation functional relative to radial measure, then adding

$$\frac{\chi(\nu)}{s}\,\delta_{\mathrm{rad}}$$

preserves parabolic dimensional covariance but changes the integrated finite part. Thus scale covariance alone does not select the Mellin value.

The earlier script comment "unique iff 2ν is not an integer" refers to a different unrestricted half-line Taylor class and must not be used as the uniqueness theorem for smooth two-dimensional radial tests.

Corollary 6.4 (The remaining D01 prescription obligation)**.** *The forbidden composite is exactly Mellin-null under the prescription in Theorem 6.2. If the complete D01 graph reduction identifies this composite and the frozen prescription selects that continuation without additional contact terms, it supplies no correction to the declared constant-profile catalogue.*

The italicized identification and prescription statements remain part of the graph-specific contract. They are not proved by $M_s(0) = 0$.

No uniform domain theorem at $\alpha = \frac{1}{2}$ or $\alpha = 1$ follows from this interior-chamber Mellin identity. Continuity of a selected polynomial is not uniform meromorphic control of the joint endpoint family.

7 Conditional reduction of a complete graph

7.1 Constant profiles are local jets

A constant transverse profile means a local jet constant near the apex, with longitudinal Fourier dependence. It is not a globally square-integrable constant section of an infinite cone.

The helicity-zero tensor restriction is

$$h_{11} = h_{22} = f_0,$$

together with the longitudinal symmetric block. The individual coordinate directions h_{11}, h_{22}, and h_{12} are useful array diagnostics, but are not separately globally admissible constant helicity-zero tensor profiles on a generic cone.

A closed transverse loop has one spectral trace. Once that trace has been reduced to a function of one identified operator, it receives one coefficient c_S, not a product of one heat coefficient for each propagator.

Assumption 7.1 (Graph-specific reduction contract)**.** For a specified complete graph G, external jet class, routing, and branch, require:

(R1) **Complete graph data.** All vertices, propagator numerators, source conventions, phases, statistics signs, symmetry factors, and independent-coordinate normalizations are specified and matched to the frozen contract.

(R2) **Operator reduction.** The fully contracted transverse expression is reduced to a finite sum

$$f_{G,S}(A_S; l, p),$$

on identified channel domains, plus a remainder proved to have zero apex UV residue.

(R3) **Domain and trace legitimacy.** Differentiated intertwiners, integrations by parts, and trace rearrangements are valid on those domains. Every boundary or finite-part defect is included or proved absent in the complete permitted sum.

(R4) **UV/IR separation.** An auxiliary infrared prescription distinguishes the UV residue from massless infrared poles. A scaleless integral is not assigned its UV pole merely by discarding its infrared part.

(R5) **Meromorphic interchange.** The apex heat identity extends to the regulated spectral functions, and the transverse subtraction, longitudinal integration, cutoff removal, and Laurent-residue operation commute in the stated prescription.

Theorem 7.2 (Conditional spectral apex reduction)**.** *Under Assumption 7.1,*

$$\operatorname{Res}_{\mathrm{UV}} G_{\mathrm{ap}}(p) = \sum_S c_S \operatorname{Res}_{\mathrm{UV}} \left[\mu^{4-d} \int \frac{\mathrm{d}^{d-2} l_{\|}}{(2\pi)^{d-2}} f_{G,S}(0; l_{\|}, p) \right]. \tag{7.1}$$

The factors, multiplicities, phases, and statistics are included in $f_{G,S}$.

Proof. Retain the auxiliary proper-time and infrared cutoffs. By (R2), the complete graph is a finite sum of functions of identified channel operators, modulo a residue-free remainder. Apply Theorem 3.2 and the channel replacement of Theorem 4.1 to each regulated heat superposition. This replaces its transverse apex functional by $c_S f_{G,S}(0; l, p)$.

Condition (R3) ensures that no boundary or trace defect has been lost. Condition (R5) permits the stated integration and residue operations. The remainder has zero residue by (R2), giving (7.1). □

Zero transverse external momentum does not prove (R2). A mixed-domain bubble or a domain-changing differential operator need not be a function of one A_S.

The v6 source reports that the archived engine data discharge (R1), and that parts of (R3) were checked [2]. The supplied script listings allow inspection of the declared routing and prefactors. Without the frozen input files, their completeness and byte identity cannot be independently re-established from the v7 material alone. v8 replays the frozen engine (read-only copies, tree hashes in Section 13.6) but does not certify completeness. In all cases, (R2), (R4), and (R5) remain hypotheses.

7.2 Longitudinal UV normalization

Write

$$\varepsilon = 4 - d, \qquad D = d - 2 = 2 - \varepsilon.$$

Lemma 7.3 (Angular coefficient of a simple UV pole)**.** *Suppose an infrared-regular Euclidean longitudinal integrand has*

$$I(R\omega;p) = R^{-2}g(\theta;p) + O(R^{-2-\delta}), \qquad \delta > 0,$$

uniformly in angle, and the dimensional angular continuation is regular at $D = 2$. Then its $1/\varepsilon$ coefficient is

$$\frac{1}{(2\pi)^2}\int_0^{2\pi} g(\theta;p)\,\mathrm{d}\theta. \tag{7.2}$$

A prescribed Wick phase multiplies this coefficient separately.

Proof. Above a fixed large radius,

$$\int_{R_0}^{\infty} R^{D-1}R^{-2}\,\mathrm{d}R = \int_{R_0}^{\infty} R^{-1-\varepsilon}\,\mathrm{d}R = \frac{R_0^{-\varepsilon}}{\varepsilon}.$$

The remainder is UV integrable. At $D = 2$, the measure supplies $(2\pi)^{-2}\mathrm{d}\theta$. □

An $O(\varepsilon)$ angular or numerator correction does not change this simple-pole coefficient *provided there is no hidden higher pole, singular projection, or evanescent mixing that invalidates that premise.* It can change finite parts. Consistent dimensional reduction is not automatic preservation of a proposed boundary identity [34].

8 The declared constant-profile catalogue

8.1 Coordinate and normalization conventions

Use the independent-coordinate order

$$\begin{aligned} &0 \leftrightarrow 00, \quad 1 \leftrightarrow 01, \quad 2 \leftrightarrow 02, \quad 3 \leftrightarrow 03, \quad 4 \leftrightarrow 11,\\ &5 \leftrightarrow 12, \quad 6 \leftrightarrow 13, \quad 7 \leftrightarrow 22, \quad 8 \leftrightarrow 23, \quad 9 \leftrightarrow 33. \end{aligned}$$

For longitudinal momentum,

$$p_\mu = (q_0, 0, 0, q_3), \qquad p^\mu = (q_0, 0, 0, -q_3), \qquad P^2 = q_0^2 - q_3^2.$$

Let c_2 denote either c_2^{F} or c_2^{CR}, and put

$$A = c_0 + 2c_1 + 4c_2, \quad B = -c_0 + 2c_1, \quad W = \frac{1}{2}(c_0 + 2c_1 + c_2). \tag{8.1}$$

The hat in $\widehat{B}_G$ denotes a declared pole record. It does not remove Assumption 7.1. The historical statement that the catalogue matches the v4 spectral logs is attributed to the v6 execution record [2].

8.2 Scalar sector

Graph	Declared coefficient of $1/(4-d)$
D04	$-\dfrac{\kappa^2}{4\pi}c_0P^2$
D05a	$-\dfrac{\kappa^2}{4\pi}(c_2-c_0)P^2$
Sum	$-\dfrac{\kappa^2}{4\pi}c_2P^2$

The supplied Q02 comparison gives symmetry factors 1 for D04 and 1/2 for D05a and D12 [5, 2]. Those source values are not re-fitted to the displayed poles.

8.3 Tensor bubbles

Every entry in Table 2 is multiplied by $\kappa^2/(16\pi)$. Transposes are understood, together with the $1 \leftrightarrow 2$ images.

Table 2: Declared tensor-bubble entries. The $(5,5)$ entries were added to the printed catalogue in v6.

Graph and key	Coefficient with $\kappa^2/(16\pi)$ removed
D11: $(4,4),(4,7),(7,7)$	$\frac{1}{2}c_0P^2$
D03: $(0,0),(0,9),(9,9)$	$-c_0P^2$
D03: $(3,3)$	$4c_0P^2$
D03: $(0,4)$	$c_0(q_0^2-3q_3^2)$
D03: $(3,4)$	$4c_0q_0q_3$
D03: $(4,9)$	$-c_0(3q_0^2-q_3^2)$
D03: $(4,4),(7,7)$	$-(c_0+4c_1)P^2$
D03: $(4,7)$	$-c_0P^2$
D03: $(5,5)$	$-8c_1P^2$
D01: $(0,0),(0,9),(9,9)$	c_0P^2
D01: $(3,3)$	$-4c_0P^2$
D01: $(4,4),(7,7)$	$(c_0-4c_1+c_2)P^2$
D01: $(4,7)$	$(c_0+c_2)P^2$
D01: $(3,4)$	$-4(c_0+2c_2)q_0q_3$
D01: $(4,9)$	$Aq_0^2-Bq_3^2$
D01: $(0,4)$	$-Bq_0^2+Aq_3^2$
D01: $(5,5)$	$-8c_1P^2$

On the physical constant scalar jet,

$$B_{f_0f_0'} = B_{44} + B_{47} + B_{74} + B_{77}.$$

No extra off-diagonal multiplicity is applied to an array already differentiated in independent coordinates.

8.4 The D02 and D12 tadpoles

The nonzero D02 coordinate entries, including their transposes, are the following, with common factor $\kappa^2 W/\pi$:

$$\begin{aligned}
&(01)(01),(02)(02) : q_3^2, && (13)(13),(23)(23) : q_0^2,\\
&(12)(12) : P^2, && (11)(22) : -\tfrac{1}{2}P^2,\\
&(01)(13),(02)(23) : -q_0q_3, && (03)(11),(03)(22) : q_0q_3,\\
&(00)(11),(00)(22) : -\tfrac{1}{2}q_3^2, && (11)(33),(22)(33) : -\tfrac{1}{2}q_0^2.
\end{aligned}$$

There are five diagonal and eighteen off-diagonal entries, hence 23 nonzero coordinate entries. The declared D12 array is

$$\widehat{B}_{\mathrm{D12}} = 0.$$

These are catalogue statements, not a claim that every massless scaleless prescription has been independently audited.

8.5 Ghost and source–ghost arrays

Define

$$R = 3c_0 + 4c_1, \qquad T = 4c_0 + 3c_1, \qquad V = c_1 - c_0. \tag{8.2}$$

For rows and columns $0, 3$,

$$\widehat{B}_{\mathrm{D06}} = \frac{\mathrm{i}\kappa^2}{8\pi} RP^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{8.3}$$

For source rows $00, 03, 33$ and upper ghost columns $0, 3$,

$$\widehat{\Gamma}_{\mathrm{D13}} = \frac{\mathrm{i}\kappa^2}{16\pi} \begin{pmatrix} -2Tq_0 & 2Vq_3 \\ -Rq_3 & Rq_0 \\ -2Vq_0 & 2Tq_3 \end{pmatrix}. \tag{8.4}$$

The transverse source rows $11, 22$, for these two ghost columns, are declared zero in the v6 reduction. This is the precise source-trace datum used below.

8.6 Vector-type keys are excluded, not zero

The supplied spectral script logs and skips mixed-channel monomials. The v6 assessment locates them on:

- D01 vector-type tensor keys involving $01, 02, 13, 23$;
- D06 transverse ghost keys $11, 22$;
- D13 keys $(01)c^1, (02)c^2, (13)c^1, (23)c^2$.

Mandatory import boundary

Mixed-helicity monomials occur on vector-type keys that are excluded from the released cone catalogue, not set to zero. Any later sheet importing this catalogue must preserve this sentence. A partial same-channel sum on an excluded key is not a complete graph value.

There is a programming distinction here: the historical `continue` statement skips a *monomial*, not necessarily the whole affected key. The corrected replay in Appendix A quarantines the entire key whenever a nonzero mixed-channel pole is found.

8.7 Endpoint checks

Proposition 8.1 (Catalogue checks)**.** *Within the comparison chamber and its stated continuous endpoints:*

(i) *all displayed F entries vanish at* $\alpha = 1$*;*

(ii) *CR and F entries agree at* $\alpha = \frac{1}{2}$*;*

(iii) *the displayed rational coefficients are continuous at* $\alpha = \frac{1}{2}, \frac{2}{3}, \frac{4}{5}$*;*

(iv) *D06 and D13 are branch independent;*

(v) *D04+D05a has the scalar sum printed above.*

Proof. At $\alpha = 1$, $c_0 = c_1 = c_2^{\mathrm{F}} = 0$. At $\alpha = \frac{1}{2}$, $c_2^{\mathrm{CR}} = c_2^{\mathrm{F}}$. There is no zero denominator at the three specified openings. The ghost arrays contain no c_2, and the scalar sum is immediate. □

These checks are not uniform domain estimates and do not prove $(\mathrm{H}_{\mathrm{reg}})$.

9 Reflection-even invariant structure of the hh array

9.1 The symmetry class

Let h, h' be the longitudinal symmetric 2×2 blocks, and let f_0, f_0' be the transverse trace coordinates. Write

$$\operatorname{tr} h = \eta^{ab} h_{ab}, \qquad h \cdot h' = h^{ab} h'_{ab}, \qquad php = p^a p^b h_{ab}, \quad a, b \in \{0, 3\}.$$

The reflection-even invariant structures quadratic in p are

$$\begin{aligned} S_1 &= P^2 \operatorname{tr} h \operatorname{tr} h', & S_2 &= P^2 h \cdot h', \\ S_3 &= (php) \operatorname{tr} h' + \operatorname{tr} h(ph'p), & & \\ S_6 &= P^2(f_0 \operatorname{tr} h' + \operatorname{tr} h\, f_0'), & S_7 &= (php) f_0' + f_0(ph'p), \\ S_8 &= P^2 f_0 f_0'. & & \end{aligned} \tag{9.1}$$

Proposition 9.1 (Six-dimensional reflection-even sector)**.** *A slot-symmetric, reflection-even $O(1,1)$-invariant bilinear of the displayed fields, quadratic in p, is a unique linear combination of the six structures in* (9.1)*.*

Proof. Reflection evenness excludes a single Levi-Civita tensor; a pair reduces to metric contractions. For two longitudinal symmetric tensors, the metric contractions are generated by

$$P^2 \operatorname{tr} h \operatorname{tr} h', \quad P^2 h \cdot h', \quad (php) \operatorname{tr} h' + \operatorname{tr} h(ph'p),$$

and the symmetrized contraction $p\,hh'\,p$. For the mixed matrices $H = \eta^{-1}h$, $H' = \eta^{-1}h'$, the polarized Cayley–Hamilton identity is

$$HH' + H'H = (\operatorname{tr} H)H' + (\operatorname{tr} H')H + \big(\operatorname{tr}(HH') - \operatorname{tr} H \operatorname{tr} H'\big) I.$$

It removes the last contraction. A longitudinal tensor paired with a scalar gives S_6, S_7; two scalars give S_8. Independence follows by comparing the coefficients of the independent field coordinates and q_0, q_3. □

Remark 9.2 (Why $SO(1,1)$ alone is insufficient)*.* Without reflection evenness, pseudoscalar terms such as

$$(\epsilon^{ab} p_a h_{bc} p^c)\, f_0' + f_0 (\epsilon^{ab} p_a h_{bc}' p^c)$$

are allowed by proper longitudinal Lorentz transformations. Thus the unqualified assertion that six structures classify *all* $SO(1,1)$-invariant bilinears is too strong. The frozen metric-contraction sector used here is reflection even.

9.2 Coefficients

Write the restricted graph bilinear as

$$\widehat{B}_G(h, f_0; h', f_0') = \frac{\kappa^2}{\pi} \sum_i \beta_i^G S_i.$$

The coefficient table is

Graph	β_1	β_2	β_3	β_6	β_7	β_8
D01	$-c_0/16$	$c_0/8$	0	$-c_0/8 - c_1/4 - c_2/2$	$c_0/4 + c_2/2$	$c_0/4 - c_1/2 + c_2/4$
D02	0	0	0	W	$-W$	$-W$
D03	$c_0/16$	$-c_0/8$	0	$3c_0/8$	$-c_0/4$	$-c_0/4 - c_1/2$
D11	0	0	0	0	0	$c_0/8$
D12	0	0	0	0	0	0
Sum	0	0	0	$3(c_0 + c_1)/4$	$-c_0/2 - c_1$	$-3c_0/8 - 2c_1 - c_2/4$

Proposition 9.3 (Catalogue decomposition and branch content)**.** *The table is an exact identity of the declared coordinate arrays, for arbitrary formal c_0, c_1, c_2. For the complete hh sum:*

(i) *the purely longitudinal structures cancel;*

(ii)
$$\frac{\kappa^2}{\pi}(\beta_6 + \beta_7) = -\frac{\kappa^2 V}{4\pi};$$

(iii) *the CR-minus-F difference is*

$$\Delta\widehat{B}^{hh} = -\frac{\nu\kappa^2}{4\pi}P^2 f_0 f_0'. \tag{9.2}$$

Proof. Substitute the definitions of S_i and compare with Table 2 and the D02 entries. This is finite polynomial coefficient comparison; no operator reduction is used.

Summing the rows gives the final row of the table. Its first three coefficients vanish, and

$$\beta_6 + \beta_7 = \frac{1}{4}(c_0 - c_1).$$

Only the total β_8 depends on c_2, with coefficient $-1/4$. Since $c_2^{\mathrm{CR}} - c_2^{\mathrm{F}} = \nu$, (9.2) follows. □

For D01 alone the branch difference has additional S_6, S_7 parts. The D02 difference is

$$\frac{\nu\kappa^2}{2\pi}(S_6 - S_7 - S_8),$$

and cancels those parts. D12 is zero in the declared catalogue. Thus the complete branch comparison includes the tadpoles.

The difference (9.2) is gauge invariant on this restricted lattice because the permitted longitudinal gauge variation has $\delta f_0 = 0$. It is a two-point branch difference, not a stress tensor or a physical core selection.

10 Ghost equation and the constant-profile ST row

10.1 The ghost equation

For a symmetric source tensor Y, define the longitudinal de Donder contraction

$$\begin{aligned}(\mathsf{F}_L Y)_0 &= \frac{q_0}{2}(Y_{00} + Y_{33}) - q_3 Y_{03},\\ (\mathsf{F}_L Y)_3 &= q_0 Y_{03} - \frac{q_3}{2}(Y_{00} + Y_{33}).\end{aligned} \tag{10.1}$$

The full contraction is

$$(\mathsf{F}Y)_\rho = p^\mu Y_{\mu\rho} - \frac{1}{2}p_\rho \eta^{\mu\nu} Y_{\mu\nu},$$

so

$$(\mathsf{F}Y)_0 = (\mathsf{F}_L Y)_0 + \frac{q_0}{2}(Y_{11} + Y_{22}), \quad (\mathsf{F}Y)_3 = (\mathsf{F}_L Y)_3 + \frac{q_3}{2}(Y_{11} + Y_{22}).$$

Theorem 10.1 (Declared projected ghost identity)**.** *The arrays* (8.3)–(8.4) *satisfy*

$$\widehat{B}_{\mathrm{D06}} = -2\eta_{\parallel}\mathsf{F}_L\widehat{\Gamma}_{\mathrm{D13}}, \qquad \eta_{\parallel} = \mathrm{diag}(1,-1). \tag{10.2}$$

With the declared zero transverse D13 source rows for columns $0, 3$, $\mathsf{F} = \mathsf{F}_L$ *on this matrix.*

Proof. Since $T + V = R$, direct multiplication gives

$$\mathsf{F}_L\widehat{\Gamma}_{\mathrm{D13}} = \frac{\mathrm{i}\kappa^2}{16\pi}RP^2\begin{pmatrix}-1 & 0\\ 0 & 1\end{pmatrix}.$$

Multiplication by $-2\eta_{\parallel}$ proves the identity. The full-contraction statement uses exactly the two declared transverse source rows. □

The identification with a graph antighost equation additionally uses the source normalization, Fourier convention, and Assumption 7.1. This is not a test of the full interacting basis.

10.2 Independent-coordinate weights

The ghost is upper-indexed, and

$$\delta_c h_{\mu\nu} = p_\mu c_\nu + p_\nu c_\mu, \qquad c_\nu = \eta_{\nu\nu} c^\nu.$$

For an invariant quadratic action written in independent coordinates,

$$\frac{\partial S_{\mathrm{inv}}}{\partial h_{(mn)}} = (2 - \delta_{mn}) E^{mn}(h). \tag{10.3}$$

Thus the lower-index 03 component carries

$$2E^{03} = -2E_{03}.$$

The sign is not an optional positive multiplicity.

The supplied v4 scan varied only $w \in \{\frac{1}{2}, 1, 2\}$, so it omitted $w = -2$ [3]. This diagnoses its reported failure, without interpreting that historical residual as an anomaly.

Supplied input 10.2 (Tree normalization, not loop fitting)**.** In the frozen conventions, the tree source is

$$\frac{\partial S}{\partial K'_{(mn)}} = \mathrm{i} A_{\mathrm{GH}} (p_m c_n + p_n c_m), \qquad A_{\mathrm{GH}} = \frac{1}{2},$$

and the invariant quadratic normalization relative to the displayed E is $N = 1$. Accordingly,

$$\lambda_{\mathrm{pred}} = \frac{N}{\mathrm{i} A_{\mathrm{GH}}} = -2\mathrm{i}. \tag{10.4}$$

These values are attributed to the frozen tree data and the v6 normalization check, not derived from the loop row [4, 2].

The optional tree routine in Appendix A checks the quadratic normalization on nonzero de Donder configurations if the frozen engine is supplied. It is not represented as run here.

10.3 An explicit algebraic closure theorem

On the axisymmetric lattice write

$$h = (h_{00}, h_{03}, h_{33}, f_0), \qquad c = (c^0, c^3), \qquad p \cdot c = q_0 c^0 + q_3 c^3.$$

Use the lower-index Einstein expression

$$\begin{aligned} E_{\mu\nu}(h) = {} & P^2 h_{\mu\nu} - p_\mu p^\rho h_{\rho\nu} - p_\nu p^\rho h_{\rho\mu} + p_\mu p_\nu \operatorname{tr}_4 h \\ & + \eta_{\mu\nu} \big(p^\rho p^\sigma h_{\rho\sigma} - P^2 \operatorname{tr}_4 h \big). \end{aligned} \tag{10.5}$$

This normalization is fixed for the present row. It is not a physical tensor-core Hessian theorem.

Let

$$\begin{aligned} \mathcal{B}(c, h) &= \widehat{B}^{hh}(\delta_c h, h), \\ \mathcal{G}(c, h) &= \sum_{(mn),\rho} \widehat{\Gamma}^{\mathrm{D13}}_{(mn),\rho} c^\rho (2 - \delta_{mn}) E^{mn}(h), \end{aligned} \tag{10.6}$$

where

$$\widehat{B}^{hh} = \mathrm{D01} + \mathrm{D02} + \mathrm{D03} + \mathrm{D11} + \mathrm{D12}$$

is the declared lattice array.

Theorem 10.3 (Constant-profile hh/Kc identity)**.** *For arbitrary formal c_0, c_1, c_2,*

$$\mathcal{B}(c,h) = -\frac{\kappa^2 V}{2\pi} P^2 f_0(p\cdot c), \qquad \mathcal{G}(c,h) = \frac{\mathrm{i}\kappa^2 V}{4\pi} P^2 f_0(p\cdot c). \tag{10.7}$$

Hence

$$\mathcal{B} + \lambda_{\mathrm{pred}}\mathcal{G} = 0. \tag{10.8}$$

For CR and F with $1/2 < \alpha < 1$ and nonzero coupling, $-2\mathrm{i}$ is the unique constant λ for which $\mathcal{B} + \lambda\mathcal{G}$ vanishes as a polynomial in all the declared fields, ghosts, and momenta.

Proof. The longitudinal gauge variation satisfies

$$\delta_c f_0 = 0, \qquad \mathrm{tr}(\delta_c h) = 2(p\cdot c), \qquad p(\delta_c h)p = 2P^2(p\cdot c).$$

By Proposition 9.3, only $\beta_6 + \beta_7 = (c_0 - c_1)/4$ contributes. This proves the formula for $\mathcal{B}$.

In two longitudinal dimensions the longitudinal part of (10.5) vanishes when $f_0 = 0$. Its contributions from the transverse trace are

$$E_{00} = -2q_3^2 f_0, \quad E_{03} = -2q_0 q_3 f_0, \quad E_{33} = -2q_0^2 f_0.$$

After raising and independent-coordinate weighting they become

$$E_{00}^{\mathrm{coord}} = -2q_3^2 f_0, \quad E_{03}^{\mathrm{coord}} = 4q_0 q_3 f_0, \quad E_{33}^{\mathrm{coord}} = -2q_0^2 f_0.$$

Multiplying by (8.4) and using $R - T = V$ gives the formula for $\mathcal{G}$. Equation (10.8) follows from (10.4).

For either physical branch, $V = (\alpha - 1)/2 \neq 0$ in the open comparison chamber. The polynomial $P^2 f_0(p\cdot c)$ is not identically zero. Thus the scalar equation $-1/2 + \lambda\mathrm{i}/4 = 0$ forces $\lambda = -2\mathrm{i}$. □

This proof establishes the identity of the declared arrays without fitting λ or invoking an unavailable log. Its complete-graph interpretation remains conditional.

Corollary 10.4 (Matter row and UNIT)**.** *The declared matter row D11+D12 is transverse on this lattice. For the UNIT substitution $c_0 = c_1 = c_2 = 1$, both $\mathcal{B}$ and $\mathcal{G}$ vanish separately. UNIT therefore cannot determine λ.*

Proof. D11 contributes only S_8, and D12 is zero. For UNIT, $V = 0$, so (10.7) gives the result. □

The normalized quadratic convention is important: if $Q(h) = \frac{1}{2}\widehat{B}^{hh}(h,h)$, then $\delta Q = \widehat{B}^{hh}(\delta h, h)$. An extra factor two would change the printed convention constant.

10.4 The three identity classes

Class	Status and target
Free source-tested	The Maxwell–Friedrichs identity $C_1 G^{\pm}_{(1)} = G^{\pm}_{(0)} C_1$ is a free-domain comparison from I-B-G-S [9]. It is not counted as a cone interacting-basis test.
Constant-profile interacting catalogue	The ghost row and Theorem 10.3 are exact identities of the stated restricted arrays, with graph identification conditional. No certified map assigns them to a named member of the programme's 26 combinations.
Full interacting basis	The 34-dimensional projection and 26 Ward combinations are the O.1 target. They have not been constructed or evaluated here. No missing combination is reported as zero.

The historical UNIT expectation also needs care. An identity holding only after full loop integration need not survive restriction to $l_\perp = 0$. For rapidly decreasing F,

$$\int \mathrm{d}l\, \mathrm{d}^2k\, \partial_{k_1}(k_1 F(l,k)) = 0,$$

but its restriction at $k = 0$ is $F(l,0)$, whose longitudinal integral need not vanish. Theorem 10.3 avoids this shortcut by direct algebra.

11 Heat moments, one-point fields, and missing two-point data

11.1 The object ledger

Object	Definition or source	What it does not establish
Apex heat moment	$\mathcal{C}_b, \mathcal{D}_b$, Theorems 3.2 and 5.2	A stress tensor, Landauer heat, or complete differentiated graph.
One-point field observable	Friedrichs Wick square, stress, and current in I-B-VP/I-B-VP-1 [11, 12]	A massless loop UV/IR theorem or two-point normalization.
Two-point graph pole	A graph satisfying (R1)–(R5)	A physical core selection or an Einstein source law.
Two-point branch difference	Equation (9.2)	A one-point stress or a tensor-core limit.

11.2 Constant-profile restriction is not injective

Proposition 11.1 (A transverse-jet kernel)**.** *On a smooth radial external-jet class containing constant and quadratic transverse jets, restriction of apex bilinear functionals to constant profiles need not be injective.*

Proof. Consider

$$L(F,G) = \int_{\mathcal{W}} (F\Delta_\perp G + G\Delta_\perp F)\, \mathrm{d}^2 y.$$

It vanishes for two constant transverse jets. For F constant near the apex and $G = r^2 g(y)$, $\Delta_\perp r^2 = 4$, so suitable longitudinal tests give a nonzero value. Thus the restriction has a nontrivial kernel. □

This does not declare a complete physical counterterm basis. It proves that the displayed restriction cannot determine every transverse-derivative coefficient.

11.3 The second scalar value is partial

For a quadratic jet, (5.3) gives a weighted spectral moment only after an operator reduction has justified that insertion. Within that reduction:

- the supplied D05a weighted longitudinal expression has large-radius behavior $O(R^{-4})$ in $D = 2$, hence no logarithmic UV term under Lemma 7.3;
- D04 requires mixed-channel traces with transverse position operators between different channel realizations. It is not determined by a function of one A_S.

Accordingly the retained status is

$$\text{D05a part} = 0 \text{ in the stated reduced model}, \qquad \text{D04 part} = \mathsf{OPEN}.$$

The second scalar basis value is not identified with its D05a part. Neither the weighted heat theorem nor the jet/curvature language of the tensor-core comparison computes D04. v8 status: still OPEN; see Section 15 and Lemma 16.3 for the ingredients supplied.

11.4 Evanescent data and the engine map

The simple-pole observation after Lemma 7.3 is a pole-level statement under its stated hypotheses. The $\tilde{g}$-leg arrays needed to determine evanescent coefficients are not supplied. Those coefficients remain open. v8 computes them under a declared slice-DRED prescription (Section 17); that is CONDITIONAL, not an identification with the frozen export.

The six reflection-even structures classify the executed constant-profile restriction. They do not supply its convention map into the programme's 18-dimensional engine basis, which also sees transverse derivatives and non-axisymmetric data. That map and the unseen components are open, not zero. v8 supplies the six constant-profile coordinates of the *hh* block exactly (Section 15); twelve remain open.

11.5 What an exact span certificate requires

The 34-dimensional basis and associated counts are retained as supplied metadata, not reconstructed from the coordinate table. A complete certificate needs the declared external-jet class, the full matrix, its coefficient field, column completeness, and the appropriate relation quotient. v8 supplies these for the declared inventory at named angles and branches (Section 14).

Lemma 11.2 (What modular rank can certify)**.** *Let M be a specified rational matrix, and let p avoid all denominators. Then*

$$\operatorname{rank}_{\mathbb{F}_p}(M \bmod p) \le \operatorname{rank}_{\mathbb{Q}} M.$$

A nonzero modular minor is an exact certificate that the corresponding rational minor is nonzero. If this lower bound equals an independently established upper bound, it certifies the exact rational rank.

Proof. Clear denominators by an integer factor not divisible by p. An integer determinant nonzero modulo p is nonzero as an integer. Apply this to a maximal nonzero modular minor. □

Thus modular arithmetic is not intrinsically incapable of an exact certificate. The unresolved issue here is the absence of the complete specified matrix and its completeness and upper-rank arguments, not a general prohibition on modular methods. For rational-function matrices, a good specialization can prove generic nonvanishing, but does not settle exceptional parameter values without additional analysis.

I-B-G's compact-test exactness warning is a useful free-field comparison: an identity on one spectral test domain is not automatically exactness on another test space [8, `rem:compact-exact`]. That comparison is not a proof of a loop span statement.

12 The continuum–preparation–instrument boundary

12.1 Three objects and three targets

Any proposed finite report of an apex coefficient must separately specify

$$\boxed{\text{defined continuum coefficient}} \longrightarrow \boxed{\text{acquisition or supplied preparation}} \longrightarrow \boxed{\text{complete finite instrument}}. \tag{12.1}$$

The arrows are obligations, not constructions supplied by the diagram. This paper studies the first object. A program printing a rational expression is not an apparatus acquiring an unknown continuum amplitude.

Following the distinction recorded in the supplied Oai290-A companion, the reporting target must also be typed [22]:

(i) recovery of an arbitrary unknown quantum code, including references;

(ii) reconstruction or estimation of one named observable;

(iii) consistency with a separately declared geometric model.

A supplied finite report of $\mathcal{D}_b$ is not automatically even an acquisition theorem for target (ii), and is not target (i). No external catalogue family supplies an apex graph premise.

The no-acquisition warning attributed to X-J-ST-2, and the window/report distinction in I-B-ECHO, preserve precisely this boundary [21, 17].

12.2 The complete destructive instrument

Let Q be a finite measured register with orthonormal basis $\{|\tau_j\rangle\}$, and let A contain every retained quantum register. Define

$$D_j = \langle\tau_j|_Q \otimes I_A, \qquad \mathcal{M}(X) = \sum_j |j\rangle\langle j|_R \otimes D_j X D_j^\dagger.$$

Proposition 12.1 (Record and retained residual)**.** *The map $\mathcal{M}$ is completely positive and trace preserving, and*

$$\operatorname{Tr}_R \mathcal{M}(X) = \operatorname{Tr}_Q X.$$

These identities hold with every untouched finite reference. The output contains A, not a returned quantum Q-factor. Positive-probability branches retain their actual normalized conditional states; zero branches have no assigned normalized state.

Proof. The recorded Kraus operators are $|j\rangle_R \otimes D_j$, and

$$\sum_j D_j^\dagger D_j = I_{QA}.$$

Writing $X = \sum_{j,k} |\tau_j\rangle\langle\tau_k| \otimes X_{jk}$ gives output blocks X_{jj}, whose sum is $\operatorname{Tr}_Q X$. The same expansion includes a reference. □

This is the complete I-B/En09-A instrument, restated locally [7, 19]. Consumption refers to the declared interface, not physical annihilation of all apparatus degrees of freedom.

12.3 I-C A and D: a classical report is not code recovery

For channels use half-diamond distance

$$d_\diamond(\Phi, \Psi) = \frac{1}{2}\|\Phi - \Psi\|_\diamond, \qquad e_{\mathrm{rec}}(\mathcal{C}) = \inf_{\mathcal{D}} d_\diamond(\mathcal{D}\mathcal{C}, \mathrm{id}).$$

Theorem 12.2 (Sharp record-only obstruction)**.** *Let $C \simeq \mathbb{C}^d$, $d \geq 2$, and suppose the complete accessible output is classical, with no input-correlated quantum side resource available to the decoder:*

$$\mathcal{C}(X) = \sum_j \operatorname{Tr}(F_j X)|j\rangle\langle j|, \qquad F_j \geq 0, \quad \sum_j F_j = I_C.$$

For every deterministic CPTP decoder $\mathcal{D}$,

$$F_e(\mathcal{D}\mathcal{C}) \leq \frac{1}{d}, \qquad d_\diamond(\mathcal{D}\mathcal{C}, \mathrm{id}_C) \geq 1 - \frac{1}{d}. \tag{12.2}$$

The distance bound is sharp over this class. If the implemented complete channel is within half-diamond distance δ of such a classical channel, its optimal recovery error is at least $\max\{0, 1 - d^{-1} - \delta\}$.

Proof. The decoded channel has measure-and-prepare form

$$\mathcal{D}\mathcal{C}(X) = \sum_j \operatorname{Tr}(F_j X)\sigma_j.$$

Its normalized Bell-input state is separable. For product density operators a, b,

$$\operatorname{Tr}[\Phi_d(a \otimes b)] = \frac{1}{d}\operatorname{Tr}(a^{\mathsf{T}} b) \leq \frac{1}{d}.$$

Convexity gives the fidelity bound. Testing the Bell projector gives the trace-distance lower bound, and the diamond distance dominates this referenced test.

For sharpness, computational dephasing is

$$\Delta_d = \frac{1}{d}\sum_{k=0}^{d-1} \mathrm{Ad}_{Z^k}, \quad Z = \sum_{j=0}^{d-1} e^{2\pi \mathrm{i} j/d}|j\rangle\langle j|.$$

Separating its identity term gives the matching upper bound $1 - 1/d$. The approximate statement follows from contraction under the decoder and the triangle inequality. □

This is the entanglement-breaking mechanism [35]; its operational forms are recorded in I-C, En07-A/B/C, and the supplied Hawk05 sources [18, 20, 24]. Appending classical heat labels or classical postprocessing does not change the premise. A retained input-dependent quantum register does change it.

The distinct Haar-average pure-state infidelity bound is $(d-1)/(d+1)$, using $F_{\mathrm{av}} = (dF_e + 1)/(d+1)$ [36]. It must not replace the whole-code diamond constant.

12.4 I-C B and F: logs are records, not a Page history

For a complete retained classical history,

$$H(Y_{\le n+1}) = H(Y_{\le n}) + H(Y_{n+1} \mid Y_{\le n}) \ge H(Y_{\le n}).$$

For a classical–quantum output,

$$S(YK) = H(Y) + \sum_y p_y S(\rho_{K|y}).$$

If only the classical record survives, its accessible observable algebra is commutative.

These are the relevant I-C B/F distinctions [18]. A particular list of SHA-256 digests is deterministic integrity data unless a probability model for that list is separately introduced. It is not an entropy of radiation, and an append-only execution transcript is not a Page history of the loop.

The finite Page means in the Hawk05 sources have fixed-input ensemble quantifiers. Whole-code recovery has arbitrary referenced-input quantifiers. No Page mean is used to infer (12.2), a Ward identity, or a gravitational endpoint [24].

12.5 I-C C: spectral heat is not thermodynamic heat

The symbols $\mathcal{C}_b$ and $\mathcal{D}_b$ denote spectral functionals. They are not Landauer costs.

If a complete non-bath system M, a fixed bath Hamiltonian H_B, inverse temperature $\beta > 0$, an initially independent Gibbs bath τ_B, and a joint unitary are separately declared, then

$$\beta Q_B = S_{\mathrm{n}}(\rho_M) - S_{\mathrm{n}}(\omega_M) + I_{\mathrm{n}}(M:B)_\omega + D_{\mathrm{n}}(\omega_B \| \tau_B). \tag{12.3}$$

Here entropies use natural logarithms and $Q_B = \operatorname{Tr} H_B(\omega_B - \tau_B)$. The equality follows from unitary entropy conservation and the definition of Gibbs relative entropy [37]. It does not require an entropy-decreasing reset merely to be valid. A strictly positive entropy decrease of the complete non-bath system implies positive heat because the last two terms are nonnegative.

No such bath, Hamiltonian, or reset is declared in this loop calculation. Thus neither the equality's physical hypotheses nor a positive reset cost is supplied by an apex heat coefficient.

12.6 I-C E: a destructive prefix selects no apex domain

A finite destructive prefix does not choose between CR, F, a positive tensor-core family, or a different boundary model. These are operator-domain specifications. Theorem 10.3 closing on both branches is not a physical branch-selection rule.

Likewise, fixed-resolution neutral-core report convergence does not imply equality or non-injectivity of exact continuum functionals. The I-C/I-B-ECHO comparison concerns fixed reports, specified continuous preparations, and complete finite instruments [18, 17].

Readout release boundary

Only the first object in (12.1) is an analytical target of this paper. A shredder instrument neither acquires the missing coefficient nor selects its domain. An echo window is not an apex residue. No readout, entropy, or thermal comparison discharges O.1.

13 Static debugging and artifact integrity

13.1 Corrections identified from the supplied listings

Issue	Correction in this revision
Six "$SO(1,1)$" structures	Require reflection evenness; otherwise Levi-Civita pseudoscalars are allowed.
Unique scale-covariant D01 extension	Replace by uniqueness of the specified meromorphic continuation. Parabolic covariance alone allows endpoint contact terms; see Remark 6.3.
Mellin value established by a routine name	Replace with the explicit formula and proof (6.2).
Mixed monomial skipped by `continue`	Quarantine the entire affected physical-branch key. Keep a coverage status and the nonzero mixed terms.
UNIT mixed terms	A genuine UNIT momentum-slice replay must retain all channel pairs. On the released lattice, where the reported mixed terms are absent, the algebraic UNIT closure is unchanged.
Missing log key interpreted as computed zero	Require an evaluated-key/coverage ledger. Legacy absence supports at most a comparison with a declared zero, not proof that the key was evaluated.
`weighted_checks.py` uses numerical `nsimplify` for an "exact" check	Use exact arithmetic in a cyclotomic quotient. Rational reconstruction from a numerical approximation is not an exact identity proof by itself.
Printed diagnostics without assertions	New acceptance checks raise exceptions on mismatch. A printed `ALL_MATCH` is not a substitute for an acceptance predicate and process status.
`eval` on serialized component keys	Use literal key parsing and a restricted arithmetic-expression parser. This is an input-integrity improvement, not a graph proof.
"Radial" Ward scan	The historical point $(X, Y) = (\mathtt{rr}, 0.2)$ has $r = \sqrt{\mathtt{rr}^2 + 0.04} \geq 0.2$, so it does not approach the apex. A true radial scan must scale both coordinates at fixed angle. No apex asymptotic theorem is inferred from that scan.
Finite resonance or random-rank scans	Distinguish exact arithmetic on a specified finite matrix from proof of candidate completeness and polynomial identities on the full jet class.
Modular rank described as only heuristic	Apply Lemma 11.2; exactness fails here because the complete matrix/certificate is absent, not because modular minors can never certify rank.
Memory-qualified C_4 normalization	Use the explicit relation in Remark 5.5.

Numerical quadrature without interval or rigorous truncation bounds remains corroboration, not a certified error enclosure. Converting high-precision numbers to ordinary Python `float` or `complex` also limits the precision actually retained.

13.2 v8 execution of the v7 program and response to its corrections

The listing of Appendix A was extracted byte for byte as `code/v8/cone_v7.py` and run unchanged. The results were:

- `selftest`: return code 0. All assertions passed, covering the cyclotomic orbifold checks for $N = 2, \ldots, 8$, the catalogue algebra, the ST row, and rank 6 for the six-structure decomposition.
- legacy log comparison: PASS on all 18 graph–branch pairs. Keys absent from the legacy logs are listed rather than read as zero.
- tree normalization: $N = 1$, $A_{\mathrm{GH}} = 1/2$, $\lambda_{\mathrm{pred}} = -2\mathrm{i}$.
- frozen-engine replay: 27 runs (graph $\times$ {CR, F, UNIT}), all with return code 0. The coverage comparison passed on all 18 physical pairs, and the physical branch values differ from `logs/v4` by exactly zero.

The replay quarantines mixed keys. These are D01 $(1,1)$, $(1,6)$, $(2,2)$, $(2,8)$, $(6,1)$, $(6,6)$, $(8,2)$, $(8,8)$; D06 $(1,1)$, $(2,2)$; and D13 $(1,1)$, $(2,2)$, $(6,1)$, $(8,2)$. On the UNIT lattice, those keys gain the mixed-pair terms that the historical `continue` discarded. For example, D01 $(1,1)$ gains $-\kappa^2 q_3^2/\pi$, D06 $(1,1)$ gains $\mathrm{i}\kappa^2 P^2/(2\pi)$, and D13 $(1,1)$ gains $\mathrm{i}\kappa^2 q_0/(4\pi)$. Table 4 gives the point-by-point verdicts.

Table 4: Verdicts on the v7 static-debugging corrections (executed in v8).

#	v7 correction	Verdict	Executed check and effect on v6 numbers
1	Six structures classify only the reflection-even sector	wording, agreed	The rank-6 assertion passes. No number changes.
2	Contact term and uniqueness of extension	wording, agreed	No number changes.
3	Mellin value should not rest on a routine name	agreed	The closed form (6.2) is checked exactly; the value is unchanged.
4	Mixed monomial skipped by `continue`	real defect (`spectral_apex.py`, line 134)	No historical physical value was affected, because the v4 logs hold no partial values on mixed keys. v8 strengthens the point: the quarantine must test pair densities, not pole totals (Section 16).
5	UNIT records omit mixed terms	real defect	The UNIT records change on 14 keys. The UNIT statements in v6 were lattice-only, so they are unaffected.
6	A missing key read as a computed zero	real defect	The v7 program's explicit `key_status` coverage was run on every key. No number changes.
7	`nsimplify` in `weighted_checks.py`	real defect	The exact cyclotomic check passes. No number changes.
8	Unasserted diagnostics	real defect	The v7 program asserts them, and every assertion passes. No number changes.
9	`eval` on keys	real defect (`st_scan`, `st_test_const`, `st_v6`)	A literal parser gives identical results.
10	Ward scan not at a radial point	real defect	The true radial scan was run. The geometric residual is $\leq 10^{-28}$ for all $r \geq 0.003$. The Friedrichs residual grows from 0.0039 at $r = 1$ to 0.101 at $r = 0.003$, whereas the historical reading was a plateau near 0.019. v6 printed no Ward-scan number.

#	v7 correction	Verdict	Executed check and effect
11	Finite resonance and random-rank scans	real limitation	Replaced by the complete exact certificates of Section 14.
12	Modular rank called only heuristic	addressed	Every exact rank equals the historical modular rank, and Lemma 11.2 applies.
13	C_4 normalization	algebra verified	$(\alpha/2)(\alpha^{-2}-1)(\alpha^{-2}+11)/90 = \mathcal{D}_0$ holds identically. The equation-number attribution to the source remains from memory (†).
14	Coarse channel marker	verified	`projector_check` passes on the frozen engine.
15	Tree-hash ordering	valid in principle	The locale is `C.UTF-8`, and the shell digest is reproducible. The frozen-tree digests are unchanged (Section 13.6).

v6 numbers changed: none. One v6 status changes: the *hh* branch difference becomes conditional (Section 16).

13.3 The coarse channel marker needs its own check

The historical spectral code labels a tensor propagator entry by whether its independent coordinates are longitudinal, mixed, or transverse. That rule is not a general helicity projector theorem.

For the four-dimensional de Donder numerator, the transverse trace–trace block vanishes, while the transverse trace can couple to the longitudinal scalar block. In that particular case the coarse marker can agree with the projected propagator. The corrected replay verifies

$$P_S \Pi = \Pi_S$$

against explicit coordinate projectors before using the marker. It does not export this four-dimensional check to unsupplied evanescent-index arrays.

13.4 The XVI-O1-ST sign witness

The supplied comparison attributes to XVI-O1-ST-v5 a cross-artifact witness: a typeset D01 sixth coefficient $-4/3$, a regression print $+4/3$, and a residual $(8/3)e_6$ identified there as a transcription discrepancy [23].

The arithmetic is immediate:

$$\left(\frac{4}{3} - \left(-\frac{4}{3}\right)\right) e_6 = \frac{8}{3} e_6 .$$

Its identification as a transcription rather than a computational error remains source-qualified: that manuscript's complete artifacts are not supplied here. The e_6 coordinate belongs to that comparison's basis and must not be silently identified with a present cone coordinate.

Selecting the source's preferred sign does not rerun a loop, clear every predecessor sign audit, or update XVI-O. A matching cone catalogue is not a universal cross-artifact sign certificate.

13.5 Historical hashes and new manifests

The following are *reported v6 labels*, not freshly recomputed hashes:

Tree	Reported SHA-256
XVI-P	`ec965c15ae25cd3bb9cc2b3a134e966b 19a1803c78d1fe5bf7554da2a8b3aaa6`
XVI-Q	`0df9a0d10409d31ac0c2d0a47197e3f6 e03c7ae258ec5fb775137f66cb0493c4`
XVI-R	`57187bc6825f5c6ae88633b7279c6c0b e57f7c4918fdbc525da4aaf958331edb`

The original shell tree hash depends on its filename ordering, exclusions, and command semantics. The new Python manifest routine records relative names and byte hashes without `find|xargs` filename ambiguity. Its tree digest uses a different explicit serialization and is *not* expected to equal the historical shell digest.

A hash binds supplied bytes to an expected digest. It does not prove a theorem, establish authorship, authenticate an unavailable file, or show that a log was generated by a claimed execution. Before/after equality also requires both actual manifests, not merely two printed matching strings.

13.6 v8: recomputed frozen-tree digests

The historical shell serialization was recomputed in v8 (locale `C.UTF-8`) on the read-only trees, both before any v8 run and after the last run:

Tree	When	SHA-256 (shell serialization)
XVI-P	before/after	`ec965c15ae25cd3bb9cc2b3a134e966b` `19a1803c78d1fe5bf7554da2a8b3aaa6`
XVI-Q	before/after	`0df9a0d10409d31ac0c2d0a47197e3f6` `e03c7ae258ec5fb775137f66cb0493c4`
XVI-R	before/after	`57187bc6825f5c6ae88633b7279c6c0b` `e57f7c4918fdbc525da4aaf958331edb`

They equal the reported v6 labels above, before and after (`logs/v8/tree_hashes_v8.json`). The v7 Python manifest serialization gives different digests, as v7 predicted: XVI-P `acd88f27...5b3c7` (1209 files), XVI-Q `c13b12d1...37586` (879 files), XVI-R `d8b9ed80...f4015` (457 files). No XVI-P, XVI-Q or XVI-R file was edited.

13.7 Computational scope of the embedded program

Appendix A consolidates the parts of the supplied scripts needed by this paper:

- heat-moment algebra and exact orbifold checks;
- declared catalogue construction;
- invariant decomposition, ghost equation, and ST identity;
- coverage-aware comparison with historical or new logs;
- optional frozen-engine momentum-slice replay and tree normalization;
- byte-level manifests.

The older exploratory jet/cohomology, kernel, scratch, and document-assembly scripts remain historical material in [3]; they are not copied as redundant submission machinery or claimed as completed certificates. The consolidated program is a replacement verification implementation, not a verbatim archive of every v1–v6 file. Its inclusion does not change their historical hashes.

14 v8: an exact complete span certificate

14.1 What is certified

The apex functionals of the historical inventory (`apex_jets.py`) are bilinear in finitely many jet atoms and polynomial in the external momentum. v8 ports them mechanically to exact Gaussian-rational arithmetic (`code/v8/apex_exact.py`; the port reproduces every historical modular count) and evaluates them on *all* relevant unit atom pairs over a complete 6×6 momentum grid, enough to determine every polynomial of the occurring degree. Therefore the evaluated columns are complete, and the row spaces computed are the exact row spaces.

Theorem 14.1 (Exact span certificate; executed)**.** *For each* $(\alpha, \textit{branch})$ *in Table 5, over* $\mathbb{Q}(\mathrm{i})$ *and for the functionals as implemented in the declared inventory: the gauge-invariant candidate ranks, the ST rank,* $\dim Z$, $\dim B$, $B \subset Z$ *and* $\dim H^0 = \dim Z - \dim B$ *are as displayed. Each rank is certified by a nonzero minor (hash and indices in* `apex_cert_*.json`*) and each kernel by verification on every complete column.*

Table 5: Complete exact certificates (v8).

(α, br)	N	ST rows	ST rank	$\dim Z$	$B \subset Z$	$\dim B$	$\dim H^0$	Ward	on K	s
$(3/10, \mathrm{CR})$	34=18+8+6+2	8640	26	8	yes	6	2	26	6	292
$(3/10, \mathrm{F})$	34=18+8+6+2	8640	26	8	yes	6	2	26	6	302
$(7071/10^4, \mathrm{CR})$	34=18+8+6+2	8640	26	8	yes	6	2	26	6	316
$(7071/10^4, \mathrm{F})$	38=21+8+7+2	14400	29	9	yes	6	3	29	6	554

Here "Ward" is the number $N - \dim Z$ of Ward-constrained combinations; at the 34-candidate points this is the v6 count 26, now exact. The ghost-number $+1$ cohomology vanishes at the sampled exact point $(7071/10^4, \mathrm{CR})$ (26/26 images closed; $H^1 = 0$).

14.2 The count 34 is angle and branch dependent

The historical modular inventory, which the exact port reproduces wherever both exist, gives:

α	branch	N	$\dim Z$	$\dim B$	$\dim H^0$	$N - \dim Z$
3/10	CR	34	8	6	2	26
3/10	F	34	8	6	2	26
41/100	CR	34	8	6	2	26
41/100	F	34	8	6	2	26
1/2	CR	42	9	7	2	33
1/2	F	42	9	7	2	33
309/500	CR	34	8	6	2	26
309/500	F	38	9	6	3	29
2/3	CR	35	8	6	2	27
2/3	F	39	9	6	3	30
7071/10000	CR	34	8	6	2	26
7071/10000	F	38	9	6	3	29
4/5	CR	34	8	6	2	26
4/5	F	38	9	6	3	29
8501/10000	CR	34	8	6	2	26
8501/10000	F	38	9	6	3	29

Thus $N = 34$, $\dim Z = 8$, $N - \dim Z = 26$ hold on CR at every tested non-resonant angle and on F for $\alpha < 1/2$; F at $\alpha > 1/2$ gives 38 with $\dim H^0 = 3$ (an additional F-branch singular exponent enters), and $\alpha = 1/2$, $2/3$ are resonant. The "34-dimensional basis" and "26 Ward combinations" of v6/v7 are therefore qualified statements; only the rows of Table 5 are exact.

14.3 Scope of the certificate

The certificate concerns the declared inventory and its functionals, at the listed rational angles. It does not prove a statement for general α (two points do not certify a rational function), does not identify the inventory with the complete interacting basis, and does not evaluate any graph. Within that scope it is CLOSED_PENDING_INDEPENDENT_REVIEW.

15 v8: Ward combinations, the 34- and 18-dimensional maps, and $\phi\phi$

15.1 Constant-profile support of the Ward combinations

Let K be the set of constant-profile coordinates, meaning jets with $m = 0$. Let π_K be the coordinate projection onto K, and let Ann_K be the annihilator of $\pi_K Z$ inside the span of the

K-supported functionals. In every exact run we found $|K| = 12$, with HH 6, CB 2, CK 3 and PP 1. The ST functionals supported on K span a space of dimension 6. The $m = 0$ rows alone already span Ann_K. Further, $\operatorname{rank} \pi_K Z = 6$ and $\operatorname{rank} \pi_K B = 5$. Also $\dim(Z \cap \ker \pi_K) = 2$, except at $(7071/10^4, \mathrm{F})$, where it is 3. The K-supported Ward combinations split by block support as

$$\underbrace{2}_{hh \text{ only}} + \underbrace{2}_{hh\text{+CK}} + \underbrace{2}_{\text{CB+CK}} = 6.$$

No combination is supported on CB, CK or PP alone. This decomposition is identical at every computed angle and branch.

15.2 The hh dictionary and the two exact Ward checks

The engine's constant-profile hh coordinates are related to the Wick-rotated longitudinal block by

$$x_{00} = -h_{\tau\tau}, \quad x_{03} = \pm \mathrm{i}\, h_{\tau z}, \quad x_{33} = h_{zz}, \quad f = \tfrac{1}{2}(h_{11} + h_{22}), \qquad q_0 = \pm \mathrm{i} q_\tau,\ q_3 = q_z.$$

Both sign choices give a solvable linear system, and the two give the same map. With $\Gamma_{\mathrm{sym}} = 4\Gamma$, the six engine coordinates y are

$$y = (-b_1,\, -2b_3,\, -b_2,\, b_6,\, b_7,\, -\tfrac{1}{4} b_8)$$

in units of κ^2/π, where b_j are the coefficients of the structures (9.1) (from $E = MS$ with $4b = M^{\mathsf{T}} y$, recorded in `hh_map_*.json`). The two hh-only Ward combinations are

$$W_1 = b_1 + 2b_3, \qquad W_2 = b_2.$$

Proposition 15.1 (Exact hh Ward checks; executed)**.** *In exact arithmetic, at $\alpha = 3/10$ and at $\alpha = 7071/10000$ and on both branches, the declared hh catalogue sum satisfies $W_1 = W_2 = 0$. The individual graphs do not. At $\alpha = 3/10$, for example,*

$$(W_1, W_2)^{\mathrm{D01}} = \left(-\tfrac{91}{5760},\ \tfrac{91}{2880}\right) = -(W_1, W_2)^{\mathrm{D03}},$$

and D02, D11 and D12 give zero.

So the check is not trivially satisfied: it tests a cancellation between D01 and D03. Like every catalogue statement, it is conditional on Assumption 7.1.

15.3 The remaining Ward combinations

The two hh+CK and two CB+CK combinations contain the v7 ST row and the projected ghost equation restricted to $m = 0$ (Theorems 10.1 and 10.3). This identification is CONDITIONAL, because v8 builds no CK/CB coordinate dictionary. The $26 - 6 = 20$ combinations that involve $m > 0$ jets have not been evaluated. They are OPEN, not zero.

15.4 The 18-dimensional engine map

The HH sector of the inventory has rank 18 on the 34-candidate branches, and rank 21 at $(7071/10^4, \mathrm{F})$. The dictionary above fixes the six constant-profile coordinates exactly, on all four computed branches. The other twelve involve transverse derivatives or non-axisymmetric jets, and their convention map is OPEN. The map is therefore PARTIAL: 6 of 18 coordinates.

15.5 The six exact values

Write the non-PP part of a closed class as $\sum_{i=1}^{6} e_i B_i$, using the exact basis of B stored in `apex_cert_*_vectors.json`. The hh constant-profile data give $\operatorname{rank}(\pi_{K\cap hh} B) = 4$, and the resulting system is consistent:

(α, branch)	e_1	e_3	$e_2 + 2e_5$	$e_4 + e_5$
$(3/10, \mathrm{CR})$	$7/40$	0	$7/120$	$-131/1920$
$(3/10, \mathrm{F})$	$7/40$	0	$7/120$	$63/640$
$(7071/10^4, \mathrm{CR})$	$2929/40000$	0	$-32842877/282840000$	$-78439319/4525440000$
$(7071/10^4, \mathrm{F})$	$2929/40000$	0	$-32842877/282840000$	$-136599773/1508480000$

At both angles, $e_1 = (1-\alpha)/4$. This is observed at two points, not proved for general α. The CR/F difference appears only in $e_4 + e_5$, where it equals $\nu/8$ exactly: $-1/6$ at $\alpha = 3/10$ and $2071/28284$ at $\alpha = 7071/10^4$, with $\nu = 4142/7071$. This matches the hh-sum difference $b_8^{\mathrm{CR}} - b_8^{\mathrm{F}} = -\nu/4$, which is consistent with (9.2). The coefficients e_5 and e_6 are not determined. They need CK/CB constant data, which in turn need a dictionary, and $m > 0$ data, because $\dim(Z \cap \ker \pi_K) = 2$ (and 3 at $(7071/10^4, \mathrm{F})$). The status is PARTIAL, and only in the stated basis.

15.6 The $\phi\phi$ values

The first $\phi\phi$ value is the catalogue PP coefficient $-c_2\kappa^2 P^2/(4\pi)$ on each branch. Its branch difference is $-\nu\kappa^2 P^2/(4\pi)$, and the engine coordinate is $y_{PP} = \kappa^2 c_2/(4\pi)$. For the second value, the D05a part is 0 in the reduced model of Section 11. The D04 part is still OPEN, for the following reasons.

- At constant profile, the replay of D04 has a single key $(0, 0)$. It has no hidden mixed pair and no evanescent coefficient, and its UNIT value is $-\kappa^2 P^2/(4\pi)$.
- The quadratic-profile part needs the r^2-weighted mixed-channel trace. v8 proves only the ingredients for it (Lemma 16.3); the mode sum and the r^2 assembly were not done.

16 v8: mixed channels, hidden pairs, and a Duhamel lemma

16.1 Feynman densities of channel pairs

For a bubble whose two lines carry channels s and t, let x be the Schwinger fraction of the t-line. The identity

$$\frac{1}{l^2(l-p)^2} = \int_0^1 \frac{\mathrm{d}x}{\big((l-xp)^2 + x(1-x)P^2\big)^2}$$

turns the pole of the slice numerator into a polynomial density $\rho_{(s,t)}(x)$. By construction, $\int_0^1 \rho_{(s,t)}\,\mathrm{d}x$ equals the circle pole of that pair. This identity was checked exactly for every bubble pair of every graph (`slice_D*.json`, field `rho_integrates_to_circle`). The cone value of the pair is

$$\int_0^1 \rho_{(s,t)}(x)\, C_{s,t}(x)\,\mathrm{d}x, \qquad C_{s,t}(x) = \mathrm{Tr}_{\mathrm{ap}}\big[e^{-(1-x)A_s} e^{-xA_t}\big].$$

For $s = t$ this is the circle pole times $\mathrm{Tr}_{\mathrm{ap}}\, e^{-A_s}$.

Definition 16.1 (Hidden mixed pair)**.** A pair with $s \neq t$ is *hidden* if its circle pole vanishes but $\rho_{(s,t)} \not\equiv 0$.

A selection rule based on totals, circle $\neq 0$, never sees a hidden pair. The historical lattice records value each pair as its circle pole times one channel coefficient, so a hidden pair receives the value zero.

16.2 The complete list (executed)

The slice run `logs/v8/slice/hidden_mixed.json` found hidden pairs in exactly two graphs.

- **D01, 13 keys.** Eight of them, $(1,1)$, $(1,6)$, $(2,2)$, $(2,8)$, $(6,1)$, $(6,6)$, $(8,2)$, $(8,8)$, were already quarantined; their pairs $(0,1)$ and $(1,0)$ have $\rho \propto (2x-1)$. The other five are **new**: $(4,4)$, $(4,7)$, $(5,5)$, $(7,4)$, $(7,7)$. Their pairs $(0,2)$ and $(2,0)$ have

$$\rho^{(4,4)}_{(0,2)} = -\frac{\mathrm{i}(q_0^2-q_3^2)(6x^2-10x+3)}{8\pi}, \qquad \rho^{(4,4)}_{(2,0)} = -\frac{\mathrm{i}(q_0^2-q_3^2)(6x^2-2x-1)}{8\pi}.$$

  The $(5,5)$ densities are 4 times these, $(7,7)$ equals $(4,4)$, and $(4,7)$, $(7,4)$ carry the opposite sign.

- **D03, 8 keys.** These are the vector-type keys, already excluded.

D02, D04, D05a, D06, D11, D12 and D13 have no hidden pairs.

Proposition 16.2 (Status of the v6 branch difference; executed and proved)**.** *The new D01 keys enter the hh sum, because the indices 4, 5 and 7 are the transverse components 11, 12 and 22. Since $\rho_{(2,0)}(1-x) = \rho_{(0,2)}(x)$ and $C_{2,0}(x) = C_{0,2}(1-x)$ by cyclicity, their total cone value is*

$$2\int_0^1 \rho_{(0,2)}(x)\, C_{0,2}(x)\,\mathrm{d}x = -2\int_0^1 R(x)\, C'_{0,2}(x)\,\mathrm{d}x.$$

Here $R(x) = \int_0^x \rho_{(0,2)}$, which is proportional to $x(x-1)(2x-3)$. This total is generically nonzero, and on the CR and F branches it is built from different channel spectra. Therefore the printed v6 value $-\nu\kappa^2 P^2 f_0 f_0'/(4\pi)$ is unchanged as the lattice evaluation of the declared catalogue. As a statement about the cone graph, however, it is CONDITIONAL *on the vanishing, or the separate evaluation, of this mixed-channel term.*

Proof. The symmetry of the densities is read off the executed list. The cyclicity follows from the cyclicity of the regulated apex trace. Integration by parts uses $R(0) = R(1) = 0$, which holds because $\int_0^1 \rho_{(0,2)} = 0$. □

16.3 A per-mode Duhamel identity

Lemma 16.3 (Duhamel identity for Bessel channels)**.** *In one angular mode, let $A_\mu = -\partial_r^2 - r^{-1}\partial_r + \mu^2 r^{-2}$ on its Friedrichs domain, with $\mu, \nu > 0$. With the bulk-subtracted mode trace,*

$$(\mu^2-\nu^2)\int_0^1 \mathrm{Tr}\big[e^{-xA_\nu}\, r^{-2}\, e^{-(1-x)A_\mu}\big]\mathrm{d}x = \frac{\mu-\nu}{2},$$

and $C'_{\mu,\nu}(x) = (\mu^2-\nu^2)\,\mathrm{Tr}[e^{-(1-x)A_\mu} r^{-2} e^{-xA_\nu}]$. In particular $C(0) = \mathrm{Tr}\, e^{-A_\mu}$ and $C(1) = \mathrm{Tr}\, e^{-A_\nu}$ after the same subtraction.

Proof. Differentiate $x \mapsto e^{-(1-x)A_\mu} e^{-xA_\nu}$ and use $A_\mu - A_\nu = (\mu^2-\nu^2)r^{-2}$. Integrate over $x \in [0,1]$, and use the per-mode value $-\mu/2$ (up to a μ-independent constant) of the subtracted trace of e^{-A_μ}. □

Numerical corroboration. We used Richardson extrapolation in the radial cutoff. For $\mu - \nu = 1$ the result was 0.49993 against 0.5, and for $\mu - \nu = 0.85$ it was 0.84990 against 0.85 (`duhamel_mode_check2.json`). This is corroboration, not an enclosure.

In the Cartesian frame, all channels share the angular phase set, and the Bessel orders are $|k/\alpha + S|$. The CR branch flips the $k = -1$ mode of $S = \pm 2$. Lemma 16.3 reduces $C'_{0,2}$ to a mode sum of r^{-2}-sandwiched traces. v8 does *not* evaluate that mode sum, and does not do the r^2-weighted D04 assembly. Both remain OPEN.

17 v8: evanescent coefficients under a declared prescription

The frozen $\tilde{g}$-leg arrays are still not supplied. v8 therefore declares a prescription and computes under it.

Definition 17.1 (Slice-DRED prescription)**.** On the zero-transverse-momentum slice, the UV degree -2 angular average uses $\hat{g}_L = \eta_L - \tilde{G}$. The symbols $\tilde{G}_{00}, \tilde{G}_{03}, \tilde{G}_{33}$ are formal, and $d = 2$ at pole level. The average is $\langle P \rangle = (l^2)^j \hat{\Delta}^j P/(2^j j! \prod_i (d+2i))$. The *evanescent coefficients* of a key are the coefficients of the $\tilde{G}$-dependent part of its pole.

Two checks were run for every pair of every graph. The $\tilde{G} = 0$ part reproduces the circle pole of the replay, and the bubble densities integrate to it. After combining with c_s and excluding quarantined and evanescent-mixed keys, the counts are as follows (`evanescent_combined.json`).

	D01	D02	D03	D04	D05a	D06	D11	D12	D13	total
CR	21	0	26	0	0	4	25	0	10	86
F	21	0	26	0	0	4	25	0	10	86
UNIT	34	0	34	0	0	6	25	0	14	113

Beyond the earlier quarantine, D01 also excludes its five evanescent-mixed keys. On the CR branch, D03, D06, D11 and D13 carry an overall factor $(\alpha - 1)$. For example,

$$\mathrm{D13}\,(4,0): \quad \frac{\mathrm{i}\kappa^2(\alpha^2-1)}{192\pi\alpha}(\tilde{G}_{00}q_0 - 2\tilde{G}_{03}q_3 + \tilde{G}_{33}q_0),$$

so the coefficient is $-c_0/16$ times the bracket. **Status:** CONDITIONAL on the declared prescription. These values are not identified with the frozen engine's $\tilde{g}$ export. Agreement at $\tilde{G} = 0$ is a consistency check, not a proof of identity preservation.

18 Open obligations and the exact release

Table 6: Open obligations after v8. Each status is stated exactly; a missing value is never zero.

Obligation	v8 status (evidence)	What is still missing
26 Ward combinations	PARTIAL. The count $N - \dim Z = 26$ is exact at the 34-candidate points (Theorem 14.1); it is 29 at $(7071/10^4, \mathrm{F})$. Six are supported at constant profile. Two hold exactly on the hh sum (Proposition 15.1); four are identified CONDITIONALly with the v7 rows.	The 20 jet-level ($m > 0$) combinations and the CK/CB dictionary.
34-dimensional projection	PARTIAL. The basis is reconstructed exactly (`basis_labels`). The projection onto the 12 constant-profile coordinates is computed, with $\operatorname{rank} \pi_K Z = 6$. The hh and PP coordinates have values. 34 is qualified by angle and branch.	The CK/CB coordinate values, all $m > 0$ coordinates, and general α.
Six exact values	PARTIAL: four combinations are fixed in the certificate's B basis, including $e_1 = (1-\alpha)/4$ and $e_3 = 0$. The CR/F difference is $\nu/8$ in $e_4 + e_5$.	e_5 and e_6. These need CK/CB data and jet data, because $\dim(Z \cap \ker \pi_K) \geq 2$.

Obligation	v8 status	Still missing
D04 part of the second $\phi\phi$ value	OPEN. The ingredients are proved: Lemma 16.3 and the reduction formula. At constant profile there is no hidden pair and no evanescent term.	The r^2-weighted mixed-channel trace and its mode sum.
Evanescent coefficients	CONDITIONAL on the slice-DRED prescription: 86 coefficients per physical branch and 113 on the UNIT lattice (Section 17).	Identification with the frozen $\tilde{g}$ export, and identity preservation.
18-dimensional engine map	PARTIAL: six of the 18 coordinates (six of 21 at $(7071/10^4, \mathrm{F})$) have an exact dictionary, and both sign choices are consistent.	Twelve transverse and non-axisymmetric coordinates.
Complete exact Span certificate	CLOSED_PENDING_INDEPENDENT_REVIEW on the finite scope of Theorem 14.1, that is, four (α, branch) points and the declared functionals.	General α, the identification of the inventory with the interacting basis, and independent review.
Vector-type keys	OPEN; excluded, not zero. D03 carries hidden mixed pairs on all eight vector keys.	Mixed-channel traces.
Hidden mixed pairs (new)	OPEN. The D01 keys $(4,4)$, $(4,7)$, $(5,5)$, $(7,4)$ and $(7,7)$ make the v6 hh branch difference CONDITIONAL (Proposition 16.2).	The value of $\int R\, C'_{0,2}$ on CR and on F.

In addition, graph identification still needs (R2), (R4), and (R5), and the undisposed parts of (R3), for every relevant graph. The D01 Mellin prescription obligation is explicit in Corollary 6.4. The general joint-apex and ($\mathrm{H}_{\mathrm{reg}}$) statements remain open.

Theorem 18.1 (Scoped v8 release)**.** *Under the displayed definitions and source qualifications:*

(i) *The normalized heat coefficients $\mathcal{C}_b^{\mathrm{F}}$ and $\mathcal{D}_b$, their specified singular-channel shifts, and the orbifold identities are established.*

(ii) *A complete graph satisfying Assumption 7.1 has the apex residue* (7.1)*.*

(iii) *The D01 composite has the local obstruction* (6.1)*; its specified Mellin continuation is* (6.2) *and is exactly zero at $z = 0$. The Mellin value does not select the graph prescription.*

(iv) *The declared coefficient catalogue satisfies the ghost identity and the constant-profile hh/Kc row. With the stated tree input, $\lambda = -2\mathrm{i}$ is fixed before loop fitting. The complete lattice branch difference is* (9.2)*.*

(v) *These rows do not determine the full interacting basis, a physical tensor-core limit, or a complete Slavnov–Taylor theorem on the cone.*

(vi) *The continuum/preparation/instrument objects in* (12.1) *remain distinct. The record-only recovery bound applies to a separately declared finite interface, not to a continuum residue. The spectral heat coefficients are not thermodynamic heat.*

(vii) *Image and echo-window functionals are not identified with the apex residue; Example 3.3 is an explicit negative control.*

(viii) *None of the obligations in Table 6 is set to zero or discharged by the companion comparisons.*

(ix) (v8, executed.) *Theorem 14.1 holds on its stated finite scope; on that scope the status is CLOSED_PENDING_INDEPENDENT_REVIEW.*

(x) (v8, executed, conditional on Assumption 7.1.) *The hh catalogue sum satisfies the two hh-supported Ward combinations exactly (Proposition 15.1); four of the six exact values are fixed in the stated basis; six of the eighteen hh engine coordinates are mapped.*

(xi) (v8.) *The v6 hh branch difference is conditional on the hidden D01 mixed-channel term of Proposition 16.2; the evanescent coefficients of Section 17 are conditional on the declared prescription.*

The release status is

O.1 on the cone = `NOT_CLOSED`, `bulk_symmetry_restored` = `false`.

Proof. Items (i)–(iii) are Theorems 3.2, 4.1, 5.2, 5.4, 7.2, 6.1, and 6.2, with Proposition 5.3 and Corollary 6.4. Item (iv) is Theorems 10.1 and 10.3, Input 10.2, and Proposition 9.3.

Proposition 11.1, the identity classification in Section 10, and the explicit missing data prove the limitation in (v). Proposition 12.1, Theorem 12.2, and the separate hypotheses of (12.3) establish (vi). Example 3.3 and the distinct definitions establish (vii). Table 6 records the remaining obligations.

Items (ix)–(xi) are Theorem 14.1, Propositions 15.1 and 16.2, and the executed records of Sections 15 and 17.

The complete O.1 requirements are not consequences of these restricted statements. The status is non-closure, not a universal no-go theorem for a future completion. □

Submission boundary

v8 adds executed evidence, and every executed statement is listed with its hash in Appendix B. The listed runs are the replay, the exact certificates, the maps, the slice and evanescent runs, and the Duhamel corroboration. v8 was compiled with pdfLaTeX. It does not claim a proof-assistant check, an independent referee review, physical acquisition, a new graph evaluation beyond the declared catalogue, or a general-α theorem from finitely many exact points. Nothing was pushed or posted.

The exact rank statement is CLOSED_PENDING_INDEPENDENT_REVIEW only on the scope of Theorem 14.1. Every other new item is PARTIAL, CONDITIONAL, or OPEN, as listed in Table 6. The present conditional statements must not be relabelled as unconditional graph theorems.

A Embedded verification and replay program

A.1 Use and acceptance conditions

Save the listing below as `cone_v7.py`. It uses Python and SymPy; numerical quadrature is not required for its exact checks. The listing is ASCII Python. It is included directly with `listings`, so typesetting needs neither shell escape nor an external code file.

Example commands:

```
python3 cone_v7.py selftest

# New logs with explicit coverage:
python3 cone_v7.py logs /path/to/logs

# Historical nonzero-only logs: comparison, not coverage proof:
python3 cone_v7.py logs /path/to/logs --legacy

# Optional replay; frozen engine is an external required input:
python3 cone_v7.py replay D01 CR \
```

```
  --engine /path/to/XVI-P/code/p5_entry --out /path/to/newlogs

python3 cone_v7.py tree \
  --engine /path/to/XVI-P/code/p5_entry

# Redirect outside the hashed tree:
python3 cone_v7.py manifest /path/to/archive > manifest.json
```

What `selftest` means. It checks the displayed finite algebraic identities. It does not independently derive the catalogue from vertices.

What `logs` means. Without `--legacy`, every compared coordinate must have an explicit evaluated-key status. With `--legacy`, missing zero entries are reported as coverage-uncertified. A successful legacy comparison is not an execution or graph certificate.

What `replay` means. It rebuilds the historical zero-transverse-momentum algebraic reduction from a supplied frozen engine. It verifies the coarse projector assignment, records all mixed pairs, and excludes an entire affected physical-branch key. For UNIT it retains all channel pairs. The output is explicitly marked `ALGEBRAIC_SLICE_ONLY`; it does not establish (R2)–(R5). The graph prefactors are declared source conventions, not a new automatic authentication of Q02.

What is not embedded. The frozen `amplitude_rules.py`, `feynman.py`, `dred.py`, vertex arrays, Q02 record, and historical JSON logs are not reproduced here. Missing dependencies fail rather than being replaced by zero.

Listing 1: Consolidated exact checker and coverage-aware optional replay.

```
#!/usr/bin/env python3
"""Scoped cone v7 checker.

Status:
  Source supplied for review; no execution is asserted by the paper.

selftest:
  Exact identities of the printed formulas, not a vertex rebuild.

logs:
  Compare explicit retained coordinates. New logs require coverage.
  --legacy permits polynomial comparison with uncertified zero coverage.

replay:
  Frozen-engine zero-transverse-momentum algebra only.
  Entire physical-branch keys with mixed poles are quarantined.
  UNIT retains all pairs. This is NOT an R2--R5 certificate.

tree:
  Check the declared tree normalization using a supplied engine.

manifest:
  Byte hashes with explicit path serialization. This is not the
  historical find/sort/xargs tree-hash algorithm.
"""
import argparse
import ast
import hashlib
import importlib
import json
import sys
import time
from pathlib import Path

import sympy as sp

I = sp.I
```

```
Q = sp.Rational
alpha = sp.Symbol("alpha", positive=True)
kappa, q0, q3 = sp.symbols("kappa q0 q3")
c0, c1, c2 = sp.symbols("c0 c1 c2")
P2 = q0**2 - q3**2
ETA = (1, -1, -1, -1)
HC = ((0, 0), (0, 1), (0, 2), (0, 3), (1, 1),
      (1, 2), (1, 3), (2, 2), (2, 3), (3, 3))
LAT = (0, 3, 9, 4, 7)
DIAGNOSTIC = (0, 3, 9, 4, 7, 5)
GRAPHS = ("D01", "D02", "D03", "D04", "D05a",
          "D06", "D11", "D12", "D13")


def zero(expr, label):
    v = sp.factor(sp.cancel(sp.expand(expr)))
    if v != 0:
        raise AssertionError(label + ": " + str(v))


def no_duplicate_pairs(pairs):
    out = {}
    for k, v in pairs:
        if k in out:
            raise ValueError("duplicate JSON key: " + str(k))
        out[k] = v
    return out


def read_json(path):
    with Path(path).open(encoding="utf-8") as f:
        return json.load(f, object_pairs_hook=no_duplicate_pairs)


def write_json(path, data):
    path = Path(path)
    path.parent.mkdir(parents=True, exist_ok=True)
    with path.open("w", encoding="utf-8") as f:
        json.dump(data, f, indent=2, sort_keys=True)
        f.write("\n")


def key_parse(text):
    obj = ast.literal_eval(text)
    if not (isinstance(obj, tuple) and len(obj) == 2
            and all(type(x) is int and 0 <= x < 10 for x in obj)):
        raise ValueError("invalid component key: " + text)
    return obj


# Restricted expression parser: no eval, attributes, function calls,
# comprehensions, or implicit imports. Serialized logs are arithmetic.
NAMES = {"alpha": alpha, "kappa": kappa, "q0": q0, "q3": q3,
         "I": I, "pi": sp.pi}


def expression(text):
    if len(text) > 100000:
        raise ValueError("expression exceeds declared input bound")

    def walk(n):
        if isinstance(n, ast.Expression):
            return walk(n.body)
        if isinstance(n, ast.Constant) and type(n.value) is int:
            return sp.Integer(n.value)
        if isinstance(n, ast.Name) and n.id in NAMES:
            return NAMES[n.id]
        if isinstance(n, ast.UnaryOp):
            if isinstance(n.op, ast.UAdd):
                return walk(n.operand)
            if isinstance(n.op, ast.USub):
                return -walk(n.operand)
        if isinstance(n, ast.BinOp):
            a, b = walk(n.left), walk(n.right)
```

```
            if isinstance(n.op, ast.Add):
                return a + b
            if isinstance(n.op, ast.Sub):
                return a - b
            if isinstance(n.op, ast.Mult):
                return a * b
            if isinstance(n.op, ast.Div):
                return a / b
            if isinstance(n.op, ast.Pow):
                if not (b.is_Integer and abs(int(b)) <= 100):
                    raise ValueError("undeclared power")
                return a**b
        raise ValueError("unsupported symbolic syntax")

    return sp.expand(walk(ast.parse(text, mode="eval")))


def C(a, b):
    return (1/a - a)/12 - b*(1-b)/(2*a)


def D(a, b):
    B2 = b*b - b + Q(1, 6)
    B4 = b**4 - 2*b**3 + b*b - Q(1, 30)
    return -11*a/180 + B2/(3*a) - B4/(6*a**3)


def physical_cs(branch):
    a = alpha
    z0 = (1-a*a)/(12*a)
    z1 = (5*a*a-6*a+1)/(12*a)
    if branch == "CR":
        z2 = (23*a*a-12*a+1)/(12*a)
    elif branch == "F":
        z2 = (23*a*a-36*a+13)/(12*a)
    elif branch == "UNIT":
        return (sp.Integer(1),)*3
    else:
        raise ValueError("unknown branch")
    return z0, z1, z2


def put(out, key, val):
    val = sp.expand(val)
    if key in out:
        zero(out[key]-val, "inconsistent duplicate " + str(key))
    out[key] = val


def symmetric(d):
    out = {}
    for (a, b), v in d.items():
        put(out, (a, b), v)
        put(out, (b, a), v)
    return out


def tensor_images(d):
    # Exchange transverse coordinates 1 and 2, then transpose slots.
    exch = {1: 2, 2: 1, 4: 7, 7: 4, 6: 8, 8: 6}
    out = dict(d)
    for (a, b), v in d.items():
        put(out, (exch.get(a, a), exch.get(b, b)), v)
    return symmetric(out)


def catalogue(cs):
    z0, z1, z2 = cs
    A, B = z0+2*z1+4*z2, -z0+2*z1
    R, T, V = 3*z0+4*z1, 4*z0+3*z1, z1-z0
    W = (z0+2*z1+z2)/2
    k16 = kappa**2/(16*sp.pi)
    out = {}
    out["D04"] = {(0, 0): -kappa**2*z0*P2/(4*sp.pi)}
```

```
    out["D05a"] = {(0, 0): -kappa**2*(z2-z0)*P2/(4*sp.pi)}
    out["D11"] = symmetric({
        (4, 4): z0*P2*k16/2,
        (4, 7): z0*P2*k16/2,
        (7, 7): z0*P2*k16/2})
    d03 = {
        (0, 0): -z0*P2, (0, 9): -z0*P2, (9, 9): -z0*P2,
        (3, 3): 4*z0*P2,
        (0, 4): z0*(q0**2-3*q3**2),
        (3, 4): 4*z0*q0*q3,
        (4, 9): -z0*(3*q0**2-q3**2),
        (4, 4): -(z0+4*z1)*P2,
        (4, 7): -z0*P2, (5, 5): -8*z1*P2}
    d01 = {
        (0, 0): z0*P2, (0, 9): z0*P2, (9, 9): z0*P2,
        (3, 3): -4*z0*P2,
        (4, 4): (z0-4*z1+z2)*P2,
        (4, 7): (z0+z2)*P2,
        (3, 4): -4*(z0+2*z2)*q0*q3,
        (4, 9): A*q0**2-B*q3**2,
        (0, 4): -B*q0**2+A*q3**2,
        (5, 5): -8*z1*P2}
    out["D03"] = {k: v*k16 for k, v in tensor_images(d03).items()}
    out["D01"] = {k: v*k16 for k, v in tensor_images(d01).items()}
    d02 = {
        (1, 1): q3**2, (2, 2): q3**2, (5, 5): P2,
        (6, 6): q0**2, (8, 8): q0**2,
        (1, 6): -q0*q3, (2, 8): -q0*q3,
        (0, 4): -q3**2/2, (0, 7): -q3**2/2,
        (3, 4): q0*q3, (3, 7): q0*q3,
        (4, 7): -P2/2,
        (4, 9): -q0**2/2, (7, 9): -q0**2/2}
    out["D02"] = {k: kappa**2*W*v/sp.pi
                  for k, v in symmetric(d02).items()}
    out["D12"] = {}
    out["D06"] = {
        (0, 0): I*kappa**2*R*P2/(8*sp.pi),
        (3, 3): I*kappa**2*R*P2/(8*sp.pi)}
    out["D13"] = {
        (0, 0): -2*T*q0, (0, 3): 2*V*q3,
        (3, 0): -R*q3, (3, 3): R*q0,
        (9, 0): -2*V*q0, (9, 3): 2*T*q3}
    out["D13"] = {k: I*kappa**2*v/(16*sp.pi)
                  for k, v in out["D13"].items()}
    return out


def invariant_data():
    x = sp.symbols("x00 x03 x33 xf")
    y = sp.symbols("y00 y03 y33 yf")

    def tr(v):
        return v[0]-v[2]

    def php(v):
        return q0**2*v[0]-2*q0*q3*v[1]+q3**2*v[2]

    dot = x[0]*y[0]-2*x[1]*y[1]+x[2]*y[2]
    S = (P2*tr(x)*tr(y), P2*dot,
         php(x)*tr(y)+tr(x)*php(y),
         P2*(x[3]*tr(y)+tr(x)*y[3]),
         php(x)*y[3]+x[3]*php(y), P2*x[3]*y[3])
    idx = {0: 0, 3: 1, 9: 2, 4: 3, 7: 3}
    return x, y, S, idx


def structure_table(cs):
    a, b, c = cs
    W = (a+2*b+c)/2
    return {
        "D01": (-a/16, a/8, 0, -a/8-b/4-c/2,
                a/4+c/2, a/4-b/2+c/4),
        "D02": (0, 0, 0, W, -W, -W),
        "D03": (a/16, -a/8, 0, 3*a/8, -a/4, -a/4-b/2),
```

```
        "D11": (0, 0, 0, 0, 0, a/8),
        "D12": (0, 0, 0, 0, 0, 0)}


def bilinear(d, x, y, idx):
    return sp.expand(sum(v*x[idx[a]]*y[idx[b]]
                         for (a, b), v in d.items()
                         if a in idx and b in idx))


def einstein_coordinates(hv):
    p = (q0, 0, 0, q3)
    pu = tuple(ETA[i]*p[i] for i in range(4))
    h = sp.zeros(4)
    for a, (m, n) in enumerate(HC):
        h[m, n] = h[n, m] = hv.get(a, 0)
    tr = sum(ETA[i]*h[i, i] for i in range(4))
    ph = [sum(pu[r]*h[r, n] for r in range(4)) for n in range(4)]
    pph = sum(pu[n]*ph[n] for n in range(4))
    E = sp.Matrix(4, 4, lambda m, n:
        P2*h[m, n]-p[m]*ph[n]-p[n]*ph[m]+p[m]*p[n]*tr
        +(ETA[m] if m == n else 0)*(pph-P2*tr))
    for n in range(4):
        zero(sum(pu[m]*E[m, n] for m in range(4)), "E transverse")
    return {a: (2-int(m == n))*ETA[m]*ETA[n]*E[m, n]
            for a, (m, n) in enumerate(HC)}


def check_catalogue_algebra():
    cat = catalogue((c0, c1, c2))
    x, y, S, idx = invariant_data()
    table = structure_table((c0, c1, c2))
    for g, coeff in table.items():
        lhs = bilinear(cat[g], x, y, idx)
        rhs = kappa**2/sp.pi*sum(a*b for a, b in zip(coeff, S))
        zero(lhs-rhs, "structure " + g)

    variables = x+y+(q0, q3)
    polys = [sp.Poly(sp.expand(s), *variables) for s in S]
    mons = sorted(set(m for p in polys for m in p.monoms()))
    matrix = sp.Matrix([[p.coeff_monomial(m) for p in polys]
                        for m in mons])
    if matrix.rank() != 6:
        raise AssertionError("six-structure independence failed")

    h00, h03, h33, f = sp.symbols("h00 h03 h33 f")
    g0, g3 = sp.symbols("ghost0 ghost3")
    hv = {0: h00, 3: h03, 9: h33, 4: f, 7: f}
    dh = {0: 2*q0*g0, 3: -q0*g3+q3*g0,
          9: -2*q3*g3, 4: 0, 7: 0}
    E = einstein_coordinates(hv)
    bterm = 0
    for g in ("D01", "D02", "D03", "D11", "D12"):
        bterm += sum(v*dh.get(a, 0)*hv.get(b, 0)
                     for (a, b), v in cat[g].items())
    gterm = sum(v*E[a]*({0: g0, 3: g3}[n])
                for (a, n), v in cat["D13"].items())
    core = kappa**2*(c1-c0)*P2*f*(q0*g0+q3*g3)/sp.pi
    zero(bterm+core/2, "hh gauge variation")
    zero(gterm-I*core/4, "antifield contraction")
    lam = 1/(I*Q(1, 2))  # Declared tree input N=1, A_GH=1/2.
    zero(lam+2*I, "a-priori lambda arithmetic")
    zero(bterm+lam*gterm, "ST row")

    # Ghost equation, separately for each upper ghost column.
    for n in (0, 3):
        Y00 = cat["D13"].get((0, n), 0)
        Y03 = cat["D13"].get((3, n), 0)
        Y33 = cat["D13"].get((9, n), 0)
        F0 = q0*(Y00+Y33)/2-q3*Y03
        F3 = q0*Y03-q3*(Y00+Y33)/2
        zero(cat["D06"].get((0, n), 0)+2*F0, "ghost row 0")
        zero(cat["D06"].get((3, n), 0)-2*F3, "ghost row 3")

```

```
    # Endpoint and branch checks are polynomial checks only.
    for branch in ("CR", "F"):
        vals = catalogue(physical_cs(branch))
        if branch == "F":
            for d in vals.values():
                for v in d.values():
                    zero(v.subs(alpha, 1), "F flat catalogue")
    cr = catalogue(physical_cs("CR"))
    fr = catalogue(physical_cs("F"))
    for g in GRAPHS:
        for key in set(cr[g]) | set(fr[g]):
            zero((cr[g].get(key, 0)-fr[g].get(key, 0))
                 .subs(alpha, Q(1, 2)), "CR=F at half")
    bhcr = sum((bilinear(cr[g], x, y, idx)
                for g in table), sp.Integer(0))
    bhf = sum((bilinear(fr[g], x, y, idx)
               for g in table), sp.Integer(0))
    nu = 2-1/alpha
    zero(bhcr-bhf+nu*kappa**2*P2*x[3]*y[3]/(4*sp.pi),
         "complete branch difference")

    if len(cat["D02"]) != 23:
        raise AssertionError("D02 entry count")
    zero(sum(table[g][0] for g in table), "longitudinal S1")
    zero(sum(table[g][1] for g in table), "longitudinal S2")
    return {"catalogue_algebra": "PASS",
            "tree_normalization": "DECLARED_INPUT_NOT_REBUILT"}


def check_heat_and_mellin():
    u = sp.Symbol("u", positive=True)
    b = sp.Symbol("b")
    T = sp.cosh((Q(1, 2)-b)*u/alpha) / (
        2*sp.sinh(u)*sp.sinh(u/(2*alpha)))
    bulk = alpha/(2*(sp.cosh(u)-1))
    cser = sp.series(T-bulk, u, 0, 2).removeO()
    zero(sp.expand(cser).coeff(u, 0)-C(alpha, b), "C coefficient")

    weighted = -2*sp.diff(T, u)/sp.sinh(u)
    wbulk = alpha/(sp.cosh(u)-1)**2
    wser = sp.expand(sp.series(weighted-wbulk, u, 0, 2).removeO())
    for k in range(1, 5):
        zero(wser.coeff(u, -k), "weighted pole cancellation")
    zero(wser.coeff(u, 0)-D(alpha, b), "D coefficient")
    nu = 2-1/alpha
    zero(D(alpha, 2*alpha)-D(alpha, 2*alpha-1)
         -2*nu*(1-nu**2)/3, "weighted flip")
    zero(D(alpha, 2*alpha).subs(alpha, 1), "weighted CR flat")

    # Exact finite checks in Q[zeta_N], not numerical nsimplify.
    x = sp.Symbol("x")
    for N in range(2, 9):
        cyclo = sp.cyclotomic_poly(N, x)
        for j in range(N):
            acc = 0
            for k in range(1, N):
                inv = sp.invert((x**k-1)**4, cyclo, x)
                phase = x**(((j+2)*k) % N) + x**(((2-j)*k) % N)
                acc = sp.rem(acc+Q(2, N)*phase*inv, cyclo, x)
            target = D(Q(1, N), Q(j, N))
            zero(sp.rem(acc-target, cyclo, x), "cyclotomic moment")

    # Finite rational algebra underlying the closed Mellin proof.
    t, n = sp.symbols("t nu")
    ratio = (t*t-n*n)/(2*t+1)
    bracket = (t+1)*ratio+n*(n+t)
    zero(bracket-t*(t+n)*(t+n+1)/(2*t+1),
         "Mellin recurrence reduction")
    # Gamma reflection gives M'_z(0)=-(nu+1)/(2s).
    return {"heat_moments": "PASS",
            "finite_cyclotomic_checks": "PASS",
            "Mellin_rational_reduction": "PASS",
            "Mellin_integral_and_continuation":
                "ANALYTICAL_PROOF_IN_PAPER"}
```

```


def retained_keys(g):
    if g in ("D04", "D05a"):
        return [(0, 0)]
    if g == "D06":
        return [(a, b) for a in (0, 3) for b in (0, 3)]
    if g == "D13":
        return [(a, b) for a in LAT for b in (0, 3)]
    if g in ("D01", "D03"):
        return [(a, b) for a in DIAGNOSTIC for b in DIAGNOSTIC]
    return [(a, b) for a in range(10) for b in range(10)]


def compare_logs(root, legacy):
    root = Path(root)
    report = {}
    for branch in ("CR", "F"):
        expected = catalogue(physical_cs(branch))
        for g in GRAPHS:
            path = root/("spec_" + g + "_" + branch + ".json")
            data = read_json(path)  # Missing files are errors.
            if data.get("graph") != g or data.get("branch") != branch:
                raise ValueError("graph/branch metadata mismatch: " + str(path))
            got = {key_parse(k): expression(v)
                   for k, v in data[
                       "Gamma_pole_coeff_of_1/(4-d)"].items()}
            status = data.get("key_status", {})
            unverified = []
            for key in retained_keys(g):
                skey = str(key)
                if not legacy:
                    st = status.get(skey)
                    if st not in ("COMPUTED_ZERO_SLICE",
                                  "COMPUTED_NONZERO_SLICE"):
                        raise ValueError("missing/excluded coverage "
                                         + str(path) + " " + skey)
                    if st == "COMPUTED_NONZERO_SLICE" and key not in got:
                        raise ValueError("nonzero status without value")
                    if st == "COMPUTED_ZERO_SLICE" and key in got:
                        zero(got[key], "zero status with nonzero value")
                elif key not in got:
                    unverified.append(skey)
                zero(got.get(key, 0)-expected[g].get(key, 0),
                     "catalogue mismatch " + g + " " + skey)
            report[g+"_"+branch] = {
                "polynomial_comparison": "PASS",
                "coverage": ("LEGACY_UNCERTIFIED" if legacy
                             else "EXPLICIT_SLICE_COVERAGE"),
                "absent_legacy_keys": unverified,
                "graph_identification": "R2_R5_NOT_CERTIFIED"}
    return report


def load_engine(folder):
    root = Path(folder).resolve(strict=True)
    for name in ("amplitude_rules.py", "feynman.py", "dred.py"):
        if not (root/name).is_file():
            raise FileNotFoundError(root/name)
    sys.path.insert(0, str(root))
    AR = importlib.import_module("amplitude_rules")
    if Path(AR.__file__).resolve().parent != root:
        raise RuntimeError("import resolved outside declared engine")
    return AR


def projector_check(AR):
    Pi = AR.pinv_sparse()
    M = sp.Matrix(10, 10, lambda a, b: sp.sympify(Pi.get((a, b), 0)))
    P0 = sp.zeros(10)
    for a in (0, 3, 9):
        P0[a, a] = 1
    for a in (4, 7):
        for b in (4, 7):
```

```
            P0[a, b] = Q(1, 2)
    P1 = sp.zeros(10)
    for a in (1, 2, 6, 8):
        P1[a, a] = 1
    P2m = sp.eye(10)-P0-P1
    projectors = (P0, P1, P2m)

    def ctype(a):
        m, n = HC[a]
        count = int(m in (0, 3))+int(n in (0, 3))
        return {2: 0, 1: 1, 0: 2}[count]

    def channel(a, b):
        ca, cb = ctype(a), ctype(b)
        if ca == cb == 1:
            return 1
        if ca == cb == 2:
            return 2
        return 0

    for s, P in enumerate(projectors):
        zero_matrix = P*P-P
        if any(sp.simplify(v) != 0 for v in zero_matrix):
            raise AssertionError("projector is not idempotent")
        PM = P*M
        for a in range(10):
            for b in range(10):
                want = M[a, b] if channel(a, b) == s else 0
                zero(PM[a, b]-want, "coarse helicity marker")
    return Pi, channel


def laurent_coefficient(expr, x, power):
    expr = sp.cancel(expr)
    num, den = sp.fraction(expr)
    dp = sp.Poly(sp.expand(den), x)
    if len(dp.terms()) != 1:
        raise ValueError("not a Laurent polynomial")
    (deg,), cf = dp.terms()[0]
    np = sp.Poly(sp.expand(num), x)
    return sp.expand(np.coeff_monomial(x**(deg+power))/cf) \
        if deg+power >= 0 else sp.Integer(0)


def replay(AR, graph, branch, outdir):
    """Historical flat momentum-slice algebra, with honest coverage."""
    start = time.time()
    F = AR.F
    if tuple(F.HC) != HC:
        raise ValueError("independent-coordinate order mismatch")
    Pi, channel = projector_check(AR)
    Gh = AR.ghost_num()
    V = F.vertices()
    kap = F.KAPPA
    LS, PS = sp.symbols("l0:4"), sp.symbols("p0:4")
    names = {**{r: LS[r] for r in range(4)},
             **{4+r: PS[r] for r in range(4)}}
    u, v = sp.symbols("u0:3"), sp.symbols("v0:3")
    slice_sub = {LS[1]: 0, LS[2]: 0, PS[1]: 0, PS[2]: 0,
                 PS[0]: q0, PS[3]: q3}

    def route(vt, routing):
        out = {}
        for key, poly in F.vt_route(vt, routing).items():
            e = poly.to_sympy(names) if hasattr(poly, "to_sympy") else poly
            # Polynomial substitution commutes with the finite contractions.
            out[key] = sp.expand(sp.sympify(e).subs(slice_sub))
        return out

    def pw(a, b, marker):
        return sp.sympify(Pi[(a, b)])*marker[channel(a, b)]

    def gw(a, b, marker):
        return sp.sympify(Gh[(a, b)])*marker[0 if a in (0, 3) else 1]
```

```
    N = {}
    def add(key, value):
        N[key] = N.get(key, 0)+value

    dens = "bubble"
    if graph == "D01":
        vt = V["V3_grav"]
        va = route(vt, [(0, 1), (-1, 0), (1, -1)])
        vb = route(vt, [(0, -1), (1, 0), (-1, 1)])
        for (a, c, e), pa in va.items():
            for (b, d, f), pb in vb.items():
                if (c, d) in Pi and (e, f) in Pi:
                    add((a, b), pa*pb*pw(c, d, u)*pw(e, f, v))
        pref = sp.sympify(AR.base_rules()["graphs"]["D01"]["S_f"]) \
            *(I*kap)**2*I**(2*vt.phase)*I**2

    elif graph == "D03":
        vt = V["V_cb_c_h"]
        va = route(vt, [(-1, 0), (1, -1), (0, 1)])
        vb = route(vt, [(-1, 1), (1, 0), (0, -1)])
        for (rho, mu, a), pa in va.items():
            for (sg, nu, b), pb in vb.items():
                if (mu, sg) in Gh and (nu, rho) in Gh:
                    add((a, b), pa*pb*gw(mu, sg, u)*gw(nu, rho, v))
        pref = -(I*kap)**2*I**(2*vt.phase)

    elif graph == "D11":
        vt = V["V_h_phi_phi"]
        va = route(vt, [(0, 1), (-1, 0), (1, -1)])
        vb = route(vt, [(0, -1), (1, 0), (-1, 1)])
        for (a, _, _), pa in va.items():
            for (b, _, _), pb in vb.items():
                add((a, b), pa*pb*u[0]*v[0])
        pref = Q(1, 2)*(I*kap)**2*I**(2*vt.phase)*I**2

    elif graph == "D04":
        vt = V["V_h_phi_phi"]
        va = route(vt, [(-1, 0), (0, 1), (1, -1)])
        vb = route(vt, [(1, 0), (0, -1), (-1, 1)])
        for (c, _, _), pa in va.items():
            for (d, _, _), pb in vb.items():
                if (c, d) in Pi:
                    add((0, 0), pa*pb*pw(c, d, u)*v[0])
        pref = (I*kap)**2*I**(2*vt.phase)*I**2

    elif graph == "D06":
        vt = V["V_cb_c_h"]
        va = route(vt, [(0, -1), (1, 0), (-1, 1)])
        vb = route(vt, [(-1, 0), (0, 1), (1, -1)])
        for (rho, mu, a), pa in va.items():
            for (sg, nu, b), pb in vb.items():
                if (mu, sg) in Gh and (a, b) in Pi:
                    add((rho, nu), pa*pb*gw(mu, sg, u)*pw(a, b, v))
        pref = (I*kap)**2*I**(2*vt.phase)*I

    elif graph == "D13":
        rules = AR.base_rules()
        vk, vg = AR.K_vertex(rules), V["V_cb_c_h"]
        va = route(vk, [(0, -1), (1, 0), (-1, 1)])
        vb = route(vg, [(-1, 0), (0, 1), (1, -1)])
        for (ap, mu, c), pa in va.items():
            for (rho, nu, d), pb in vb.items():
                if (mu, rho) in Gh and (c, d) in Pi:
                    add((ap, nu), pa*pb*gw(mu, rho, u)*pw(c, d, v))
        r = rules["graphs"]["D13"]
        pref = sp.sympify(r["ext_sign"]*r["S_f"]*r["loop_sign"]) \
            *(I*kap)**2*I**(vk.phase+vg.phase)*I

    elif graph == "D02":
        dens = "tadpole"
        vt = V["V4_grav"]
        va = route(vt, [(0, 1), (0, -1), (1, 0), (-1, 0)])
        for (a, b, c, d), pa in va.items():
```

```
            if (c, d) in Pi:
                add((a, b), pa*pw(c, d, u)*v[0])
        pref = Q(1, 2)*(I*kap**2)*I**vt.phase*I

    elif graph == "D12":
        dens = "tadpole"
        vt = V["V_hh_phi_phi"]
        va = route(vt, [(0, 1), (0, -1), (1, 0), (-1, 0)])
        for (a, b, _, _), pa in va.items():
            add((a, b), pa*u[0]*v[0])
        pref = Q(1, 2)*(I*kap**2)*I**vt.phase*I

    elif graph == "D05a":
        dens = "tadpole"
        vt = V["V_hh_phi_phi"]
        va = route(vt, [(1, 0), (-1, 0), (0, 1), (0, -1)])
        for (c, d, _, _), pa in va.items():
            if (c, d) in Pi:
                add((0, 0), pa*pw(c, d, u)*v[0])
        pref = Q(1, 2)*(I*kap**2)*I**vt.phase*I
    else:
        raise ValueError("unknown graph")

    # Exact circle average via a Laurent variable z=e^(i theta).
    z, w = sp.symbols("z w", nonzero=True)
    cosine = (z+1/z)/2
    sine = (z-1/z)/(2*I)
    lm0, lm3 = I*cosine/w, sine/w
    l2 = -1/w**2
    den = l2 if dens == "tadpole" else \
        l2*((lm0-q0)**2-(lm3-q3)**2)

    def pole(expr):
        e = sp.expand(expr.subs({LS[0]: lm0, LS[3]: lm3}))
        if e == 0:
            return sp.Integer(0)
        ser = sp.series(e/den, w, 0, 3).removeO()
        g2 = sp.expand(ser).coeff(w, 2)
        average = laurent_coefficient(g2, z, 0)
        # i/(4 pi^2) times 2 pi times the circle average.
        return sp.cancel(I*average/(2*sp.pi))

    if graph in ("D04", "D05a"):
        domain = [(0, 0)]
    elif graph == "D06":
        domain = [(a, b) for a in range(4) for b in range(4)]
    elif graph == "D13":
        domain = [(a, b) for a in range(10) for b in range(4)]
    else:
        domain = [(a, b) for a in range(10) for b in range(10)]

    cs = physical_cs(branch)
    values, states, mixed = {}, {}, {}
    for key in domain:
        e = sp.expand(N.get(key, 0))
        pair_poles = {}
        if e != 0:
            for mon, cf in sp.Poly(e, *u, *v).terms():
                iu = [i for i in range(3) if mon[i]]
                iv = [i for i in range(3) if mon[3+i]]
                if len(iu) != 1 or len(iv) != 1 \
                        or sum(mon[:3]) != 1 or sum(mon[3:]) != 1:
                    raise ValueError("unexpected channel-marker degree")
                s, t = iu[0], iv[0]
                if dens == "tadpole":
                    t = s
                pv = pole(cf)
                pair_poles[(s, t)] = pair_poles.get((s, t), 0)+pv
        pair_poles = {st: sp.factor(vv)
                      for st, vv in pair_poles.items()
                      if sp.simplify(vv) != 0}
        bad = {str(st): str(vv) for st, vv in pair_poles.items()
               if st[0] != st[1]}
        skey = str(key)
```

```
        if bad:
            mixed[skey] = bad
        if bad and branch != "UNIT":
            states[skey] = "EXCLUDED_MIXED_CHANNEL"
            continue
        total = 0
        for (s, t), pv in pair_poles.items():
            total += pv if branch == "UNIT" else pv*cs[s]
        value = sp.factor(sp.cancel(-I*pref*total))
        if value == 0:
            states[skey] = "COMPUTED_ZERO_SLICE"
        else:
            states[skey] = "COMPUTED_NONZERO_SLICE"
            values[skey] = str(value)

    result = {
        "graph": graph, "branch": branch,
        "status": "ALGEBRAIC_SLICE_ONLY",
        "prescription_status": "R2_R3_R4_R5_NOT_CERTIFIED",
        "prefactor_status": "DECLARED_FROZEN_SOURCE_CONVENTIONS",
        "pref": str(pref),
        "key_status": states,
        "mixed_channel_poles_before_prefactor": mixed,
        "Gamma_pole_coeff_of_1/(4-d)": values,
        "seconds": time.time()-start}
    path = Path(outdir)/("spec_"+graph+"_"+branch+".json")
    write_json(path, result)
    return {"written": str(path), "status": result["status"],
            "excluded_keys": sum(s == "EXCLUDED_MIXED_CHANNEL"
                                 for s in states.values())}


def tree_normalization(AR):
    F = AR.F
    if tuple(F.HC) != HC:
        raise ValueError("coordinate order mismatch")
    vt = F.grav_vertex(2)
    ks = [F.ksym(j) for j in range(2)]
    names = {4*j+r: ks[j][r] for j in range(2) for r in range(4)}
    p = (q0, 0, 0, q3)
    h00, h03, h33, f = sp.symbols("h00 h03 h33 f")
    # De Donder equations on h11=h22=f.
    F0 = q0*(h00+h33+2*f)/2-q3*h03
    F3 = q0*h03-q3*(h00+h33-2*f)/2
    solutions = sp.solve((F0, F3), (h00, h03), dict=True)
    if len(solutions) != 1:
        raise AssertionError("generic de Donder solve failed")
    sol = solutions[0]
    hv = {0: sol[h00], 3: sol[h03], 9: h33, 4: f, 7: f}
    Eu = einstein_coordinates(hv)
    ratios = []
    for a in range(10):
        val = 0
        for b in range(10):
            poly = vt.T.get((a, b))
            if poly is None:
                continue
            e = I**vt.phase*poly.to_sympy(names)
            sub = {ks[0][r]: p[r] for r in range(4)}
            sub.update({ks[1][r]: -p[r] for r in range(4)})
            val += e.subs(sub)*hv.get(b, 0)
        val = sp.cancel(val)
        den = sp.cancel(Eu[a])
        if den == 0:
            zero(val, "tree zero-coordinate comparison")
        else:
            ratios.append(sp.cancel(val/den))
    if not ratios:
        raise AssertionError("tree check had no nonzero probe")
    N = ratios[0]
    for r in ratios[1:]:
        zero(r-N, "tree normalization not constant")
    zero(N-1, "declared N")
    Agh = sp.sympify(F.A_GH)
```

```
    zero(Agh-Q(1, 2), "declared A_GH")
    zero(N/(I*Agh)+2*I, "tree-predicted lambda")
    return {"N": str(N), "A_GH": str(Agh),
            "lambda_pred": str(N/(I*Agh)),
            "scope": "TREE_NORMALIZATION_ON_DECLARED_DD_CLASS"}


def manifest(root):
    root = Path(root).resolve(strict=True)
    if not root.is_dir():
        raise ValueError("manifest root is not a directory")
    files = []
    for p in sorted(root.rglob("*"),
                    key=lambda p: p.relative_to(root).as_posix()):
        if p.is_symlink():
            raise ValueError("symlink requires a separate policy: " + str(p))
        rel = p.relative_to(root)
        if ".pytest_cache" in rel.parts or p.suffix == ".pyc":
            continue
        if p.is_file():
            digest = hashlib.sha256(p.read_bytes()).hexdigest()
            files.append((rel.as_posix(), digest))
    tree = hashlib.sha256()
    for rel, digest in files:
        name = rel.encode("utf-8")
        tree.update(len(name).to_bytes(8, "big"))
        tree.update(name)
        tree.update(bytes.fromhex(digest))
    return {"serialization":
                "sorted UTF8 relative names; 8-byte length; name; raw SHA256",
            "exclusions": [".pytest_cache path component", "*.pyc"],
            "symlinks": "REJECTED",
            "files": dict(files), "tree_sha256": tree.hexdigest()}


def main():
    p = argparse.ArgumentParser()
    sub = p.add_subparsers(dest="command", required=True)
    sub.add_parser("selftest")
    lp = sub.add_parser("logs")
    lp.add_argument("root")
    lp.add_argument("--legacy", action="store_true")
    rp = sub.add_parser("replay")
    rp.add_argument("graph", choices=GRAPHS)
    rp.add_argument("branch", choices=("CR", "F", "UNIT"))
    rp.add_argument("--engine", required=True)
    rp.add_argument("--out", required=True)
    tp = sub.add_parser("tree")
    tp.add_argument("--engine", required=True)
    mp = sub.add_parser("manifest")
    mp.add_argument("root")
    a = p.parse_args()
    if a.command == "selftest":
        result = {**check_heat_and_mellin(), **check_catalogue_algebra(),
                  "execution_claim":
                      "THIS_OUTPUT_EXISTS_ONLY_IF_THE_USER_RUNS_THE_PROGRAM",
                  "Q1": "NOT_CLOSED",
                  "bulk_symmetry_restored": False}
    elif a.command == "logs":
        result = compare_logs(a.root, a.legacy)
    elif a.command == "replay":
        result = replay(load_engine(a.engine), a.graph, a.branch, a.out)
    elif a.command == "tree":
        result = tree_normalization(load_engine(a.engine))
    else:
        result = manifest(a.root)
    print(json.dumps(result, indent=2, sort_keys=True))


if __name__ == "__main__":
    main()
```

A.2 Reproducibility obligations not replaced by this listing

A full computational submission should additionally provide:

(1) immutable frozen input files and their actual byte manifests;

(2) the graph inventory and the complete coordinate/domain coverage ledger;

(3) exact contracted spectral expressions before evaluation at zero transverse spectral parameter;

(4) the regulated trace reduction, with boundary defects;

(5) the UV/IR separation and meromorphic interchange certificate;

(6) source normalization and the complete relevant ST functional;

(7) the full basis projection and exact rank/span certificates;

(8) software versions, commands, process return codes, raw outputs, failed runs, and replay-versus-rebuild declarations.

A successful run of the embedded program would not, by itself, provide items 3–5 or 7. v8 ran it (Section 13.2) and separately supplies item 7 on a finite scope (Section 14); items 3–5 are not supplied.

B v8 artifacts, hashes and rerun commands

Paths are relative to `/workspace/XVI-01-CONE/` on the audit box. Long jobs were run detached with `setsid` and logged. The environment is Python 3.13.5, SymPy 1.14.0, with `LC_ALL=C.UTF-8`. The frozen trees were hashed before the first v8 run and after the last one; the before/after values are identical: XVI-P: yes, XVI-Q: yes, XVI-R: yes. A full manifest is `logs/v8/artifact_manifest_v8.sha256`; the TEX source and PDF are hashed in `SHA256SUMS_v8`, outside this document.

Rerun commands.

```
cd /workspace/XVI-01-CONE
python3 code/v8/cone_v7.py selftest
python3 code/v8/cone_v7.py logs logs/v4 --legacy
bash code/v8/run_replay.sh   # 27 runs
python3 code/v8/cone_v7.py logs logs/v8/replay
python3 code/v8/ward_scan_radial.py 0.85
python3 code/v8/apex_exact_run.py 7071/10000 CR
python3 code/v8/apex_certificate.py 3/10 CR      # also 3/10 F, 7071/10000 CR|F
python3 code/v8/k_support.py 3_10_CR             # per tag
python3 code/v8/hh_map.py 3/10 CR                # per (alpha, branch)
python3 code/v8/exact_values.py 3_10_CR 3_10_F 7071_10000_CR 7071_10000_F
bash code/v8/run_slice.sh && bash code/v8/run_slice2.sh
python3 code/v8/evanescent_combine.py
python3 code/v8/duhamel_mode_check2.py
python3 code/artifacts_v8.py && python3 code/build_v8.py
pdflatex -halt-on-error XVI-01-CONE-v8.tex   # twice
```

The certificate runs take 5–10 minutes each and stay below 2 GB of memory. The box limit is about 7 GB, so they were run at most three at a time.

Table 7: SHA-256 of v8 artifacts.

File	SHA-256
Base source (read-only) (1 files)	
XVI-01-CONE-v7.tex	d0ba7b9cdbf535d1296a74146011c1cb1269ff53148d3a96add64574873d5132
v8 programs (23 files)	
code/v8/apex_certificate.py	14a49b42dcb7c1fb53a69825a75a9a6dcfa36ad4f88f7f8ca80e95f47818b1ca
code/v8/apex_certificate_la.py	803c363e5aa3fd32824286bf5f4da891067c2a6e91c80705acf14178787cfb21
code/v8/apex_exact.py	ba6b0946e84e9f2fb3a60a24d38a3e4343e3887ae6c79c385b88e5c0db4ec46b
code/v8/apex_exact_run.py	db8f950dd31e012044776bddc3bd594aa2edededead73794459b71301321e23d48
code/v8/cone_v7.py	55b197552eb4c400c526cd39e4964a4cc8ec1fb89cf45dd41b1336dd42ba9b07
code/v8/duhamel_mode_check.py	b7a1183a11c4f0219b700a7d6ca0056f63aa8fc9efd3012d89a156d8310794f5
code/v8/duhamel_mode_check2.py	b010c66de54a5b7fde6bfc8efd42a4d618533f65390337ee0cdb8d250b698aaa
code/v8/evanescent_combine.py	e967018847f31c26448a832381c558fb01dde75dfb36bf60402e8873c8e58a0c
code/v8/exact_values.py	4b4ad1682afb5629f24df91b97a9a905a980ba1fa034fa060c087dfe9f7bfae3
code/v8/gq.py	a48693860dde5d3d0b244ca0b13c00cea1988849a45a3b16fe39d1cc7e1eae3b
code/v8/hh_map.py	372b3f01f084e1bb8091c933d3d979a445b445febc6610694048a7253c850cee
code/v8/k_support.py	471f02fa05515d76d4fd94b57b38811bdcafac450d0cd900e9466dfa076aa94f
code/v8/make_apex_exact.py	2f4e74c6c3ce1c0d92470f7babbb27f0b9d53515cd390270d352303cdc6da6fa
code/v8/make_apex_exact_run.py	c8e9abbb5402d35f1c97dafce62215c94d0bfae44e3ba6b266bedc23902fcb18
code/v8/slice_engine.py	296bfb8541ded14d27f1050cf8602693fb1a6f4c7c99cdd378b57eaed40069ed
code/v8/slice_run.py	2b46ef8d8c6dc8c9085725eccdf0d89aad609098ea913e57ba55f9a92b2e0688
code/v8/test_slice.py	75acc5a9c6deea06be6623038e745ddbff481bb12ae7b2a1dc23d65f4279cf70
code/v8/ward_scan_radial.py	453a8f3dfd959465b5722a710485fea655b4265af5cca914e7d09499049fafca
code/v8/run_replay.sh	c459fd5759b7cdec921d12fdd27ad3241a118c5c12361541e23b3916e07ef968
code/v8/run_slice.sh	2ff93bab2df9006a0054b71f39b4f055f475eb2c230fc9d740bec708b64e8d7b
code/v8/run_slice2.sh	bd47fcb81bdfa021554c3d667088249b3973ac8dde448239d696955778e90251
code/build_v8.py	c83d29a050937ab65347255ece236c47fc3ef8f78063f50710391532f2bf1263
code/artifacts_v8.py	adb9abcb3d879cdb7eb32e1956c16ae21ada684b5bb3aabcb0b4c8ab0f7f7766
v8 source fragments (9 files)	
v8src/abstract.tex	f5d9e2b3fd61a71d616d41562c68ba2d1bec5537f5c069b29dcaa0abb5c7d130
v8src/boundary.tex	33dd134192a6b257f9065161aba3ebf2df4238cb834ebb9dd066465c57e40fa3
v8src/opentable.tex	03a4908ff49b8f49d0ad402839aafcd70ad891e57a8a20f01b54322758fbe4bb
v8src/response.tex	7dd77765cbdc314c760cebc8eaa66839f0a4c329cb706075098a0399802d9336
v8src/sec_evan.tex	47e4623fa8bc6200026cf430642b705674154a95d40e472bc37ad95a17673422
v8src/sec_map.tex	83e8eb0f2e6b6b24ee812bd5924a0da5344c98bd34a7d091fdafed4ef2d0d1f8
v8src/sec_mixed.tex	23807ea310beb231d885e08bed37aebbe69014c601b91becbc7d08b0f57fb1ed
v8src/whatv8.tex	7d654d0008959917d3dee81e41930b6e68ad7bc058f9897b3fa6c621becc021b
v8src/bibmarks.json	12e6130eabfd85e6fc1841c039149ad8688cd32a66f934b606825675f3cc4cd3
v7-program runs (4 files)	
logs/v8/v7_logs_legacy.json	0674a32ac36a26c0a485c5591fb006bc4739e4159c6019912fb2860c37b1ca4f
logs/v8/v7_logs_replay.json	856162e1d1ba8365213ee7af80e1a661a02535b0512d011d99aa5a353d8871de
logs/v8/v7_selftest.json	5d0668fc426d17a17cb0b058f78ce919532f7003c0d8d548f0440a4bb10277ef
logs/v8/v7_tree.json	acf9699d11beaca179da32ffab88ee57366f1cbe1da8ea48aa8e7876538d5391
Replay (27 files)	
logs/v8/replay/spec_D01_CR.json	acce791a88e9204de5691ef10b411cc9809625ea5fa1cfe2afb0eb21c54b6003
logs/v8/replay/spec_D01_F.json	79a84353204e8e9e50fc91d924b4c4da91bab831d5a10e98ffe74dc118f13f03
logs/v8/replay/spec_D01_UNIT.json	5649acd95b4f21ca4a564945e3eb35cc968a051c0b6e5f29934e5384b5cbea8c
logs/v8/replay/spec_D02_CR.json	7f4a1d36f5f1723abe33324b654b07cb5e51bde054f82c7e9235c3abe5decd50
logs/v8/replay/spec_D02_F.json	99bb7ba60647bc744d2da6007717972772910937 9e857bbf3f9f8ff466035062
logs/v8/replay/spec_D02_UNIT.json	fd11e5c3f5272aa46177f9adcd2cf56fd30b31b1f28ff2f6118f6afdc5959dbb
logs/v8/replay/spec_D03_CR.json	2bec5983c7c649fe57e29452e4819847e581d9a4ff059c7df4f50127034f14c6
logs/v8/replay/spec_D03_F.json	d3f142193c21fb2bcb45bb44692cba7cd09a59131cfc2557ee084cbf7aa23abb
logs/v8/replay/spec_D03_UNIT.json	3e1e4792fb8ec91035b5ef3c9bb6343255c46d1f099c28750a56ef3044e6f24c
logs/v8/replay/spec_D04_CR.json	a2151dc64d8159bd2183dd4fb84b9fa69b2500ccb8e959282279fae9ba2d3d71
logs/v8/replay/spec_D04_F.json	542b00df420c2d5bb8cc884b8103cc9fa705d3e60a06458bcbd0ef8f5da2ae19
logs/v8/replay/spec_D04_UNIT.json	512cdf20a2a5df3a0ef7df19acd867b5c70e65a53ca3c57b34adf14b36cb7446
logs/v8/replay/spec_D05a_CR.json	179e86ad7825cfebf3a48ab4e670b639518fbab389513f9fd220bc458e734317

File	SHA-256
logs/v8/replay/spec_D05a_F.json	4f5cbe0ae2b9a9965780a290bc3652e960c356383716794b659f2549f94e12ea
logs/v8/replay/spec_D05a_UNIT.json	ca56b30462aa4fca5a73bac77870b9f55e377f43fe9cd7afe17d1ce0ecdd105f
logs/v8/replay/spec_D06_CR.json	62ef7ca5c49060ff207c284497ff1eaf60e7214227e0164e5e8fb13a6fb0d1f0
logs/v8/replay/spec_D06_F.json	43575824758f99a429e9980f88bb9441687a209a0e7f6f4b0172a039e554ed2b
logs/v8/replay/spec_D06_UNIT.json	e0967ee852a1f35946ea020ca4afc39b83f961bcae556b0ff6f890493dd97916
logs/v8/replay/spec_D11_CR.json	05f947251f8a3d1cee8c721e998a5e431c71fe87d75abb76881aac32d99698d5
logs/v8/replay/spec_D11_F.json	a350cacd428979ff494fdc2b7d0069a12a8955204203442 59fee3252bac0b0a5
logs/v8/replay/spec_D11_UNIT.json	b9c70d687aee15bd7c9f53697fb04264da32cba89e4824c0eb5d1fe1400fe30a
logs/v8/replay/spec_D12_CR.json	6675f26bf142c5e549d2d4e4ed78640f30de99bbc2c8c3d57c6b5de4a19059fa
logs/v8/replay/spec_D12_F.json	356db7ea109fe8dfce4e29c5c67bf7c3a6c525bcb0fb7cb4c3e5f66b12674ee6
logs/v8/replay/spec_D12_UNIT.json	8c3d2c8aa6018bac44281d5d21aeb77cd35c53cb3663fabe1ea920da5196f6c0
logs/v8/replay/spec_D13_CR.json	794463632e6f4f6303047f8c94b12172874b60f1dce1c5d35bdfec781c6eb249
logs/v8/replay/spec_D13_F.json	ff97041eff19ac02e6009fd36f30748715f8856eaaa5dc6fef165d82ba78342f
logs/v8/replay/spec_D13_UNIT.json	d4041679d0a9c8b27237d1746a9c72dc661bba20b8c76591f8ab7a37570b7d9b
Ward scan (2 files)	
logs/v8/ward/wardscan_radial_0.85_fried.json	c802124092dc962b8fa7ccac419650eeae2a906b78b5ab5efbdd32bfa6675258
logs/v8/ward/wardscan_radial_0.85_geom.json	007709fdd124593107898379c9c318e188c6a926d67905ff2c3908d496ef57a3
Exact certificates and maps (17 files)	
logs/v8/cert/apex_cert_3_10_CR.json	001aa482da32b879365eaad7cd97e584b587895074354840f14b0ad94df53a96
logs/v8/cert/apex_cert_3_10_CR_vectors.json	a719330531cb070bea918f2f8a0b755e48e0100974756bc2aa932e779cff9319
logs/v8/cert/apex_cert_3_10_F.json	fc76e1a9d2fb95eb2256c5ee4a24353d8efce1c9e38942a67b65ca64d979f93b
logs/v8/cert/apex_cert_3_10_F_vectors.json	a719330531cb070bea918f2f8a0b755e48e0100974756bc2aa932e779cff9319
logs/v8/cert/apex_cert_7071_10000_CR.json	204c46e1b8ac59609072a060018373f67ac19e93717fbdc56b0547b863330a0b
logs/v8/cert/apex_cert_7071_10000_CR_vectors.json	a719330531cb070bea918f2f8a0b755e48e0100974756bc2aa932e779cff9319
logs/v8/cert/apex_cert_7071_10000_F.json	f3fd2f7010379d22b7236f92247f19c8a82a1eaa9947b7c7be27e9cdcff02af0
logs/v8/cert/apex_cert_7071_10000_F_vectors.json	ef66990df199ceca552c9986688ebfe82f3c1e050bedfe76b16a415ed40dc728
logs/v8/cert/exact_values.json	21aefbf917ccda4a9a70bc0bc8a1405ea2e556713bd6bd4aa104aeed757926e7
logs/v8/cert/hh_map_3_10_CR.json	ee9d20a4b630496ba5340c74e05ae16d2613d21a1b6cb05433b21f5cc1939894
logs/v8/cert/hh_map_3_10_F.json	de28306ff18ec729efd27eca9269fa6d75a2fef8839950c4f11978ae41c1c40e
logs/v8/cert/hh_map_7071_10000_CR.json	d6c9abe28830ccd985c69466ee1eca53344b69b27216ea3e8a8ef68100aebad5
logs/v8/cert/hh_map_7071_10000_F.json	e6f492d4cf6f19b6c9b75f3be96b13ac97a0382c23f8902f92261beb9f30742e
logs/v8/cert/k_support_3_10_CR.json	0e442beaef6ed47ef2e53bfc7f7f9034a83536a4cacda009f88e0fb5fbe22a45
logs/v8/cert/k_support_3_10_F.json	0e442beaef6ed47ef2e53bfc7f7f9034a83536a4cacda009f88e0fb5fbe22a45
logs/v8/cert/k_support_7071_10000_CR.json	0e442beaef6ed47ef2e53bfc7f7f9034a83536a4cacda009f88e0fb5fbe22a45
logs/v8/cert/k_support_7071_10000_F.json	0e442beaef6ed47ef2e53bfc7f7f9034a83536a4cacda009f88e0fb5fbe22a45
Exact sampled port (1 files)	
logs/v8/apex_exact/apex_7071_10000_CR_seed0.json	9b14e0b3f879676cdcd97d162f5507ff9fb9ae64026473b76c00da0d2ad8f4d1
Slice / evanescent / hidden pairs (11 files)	
logs/v8/slice/evanescent_combined.json	de6d4bf42a5477bb9c82a331d0a5d77447dee43f8b950ee1140aaf5832ef1054
logs/v8/slice/hidden_mixed.json	4f4c0cb1d3e8a0b53adcd0c94c39515f8d7b06b64f00c043de47d8a2b8731077
logs/v8/slice/slice_D01.json	0a55a654154890d9c8c50e70638d47f59340c05b8afa87842c6e9be2b492f6fb
logs/v8/slice/slice_D02.json	06e19f1c9e8f2acd0f4546c03e3b3e8a4273456c87b5304eab5551f23ef92e74
logs/v8/slice/slice_D03.json	c7eb6a6efcfc389f9d2439f2472b62b47ef0ef6201deff6f4ef92d968acf418d
logs/v8/slice/slice_D04.json	cc23ad207c516682715544fbafe2434dce83e84785d19e8383adeab0c1a833b7
logs/v8/slice/slice_D05a.json	5a6b5c35ae58f9197a75128b7e002ebee0a8688682b0d4e8f8414687861fa897
logs/v8/slice/slice_D06.json	cf0bcdad0d20b76076fcf327fafbbe5c5f2a6a568537e98fd7d4383e64d09786
logs/v8/slice/slice_D11.json	f4afed2f0745d95d685294001dab6e0099720733a6e624df8a5971336fcd5656
logs/v8/slice/slice_D12.json	acfa367bac579ab8f18c2bd731b3b3550a83aa1862722a4da2250774cdef88be
logs/v8/slice/slice_D13.json	eeb384f033363738b9bcd5d0160ee3d69299c8426c2c2fb35ee9b62d628226c6
Duhamel corroboration (2 files)	
logs/v8/duhamel_mode_check.json	2931da69cbc9873704391c64ac13299b41666f1ad59eeaeb67ac4eab63eb86db
logs/v8/duhamel_mode_check2.json	74bc8b04bc98559da3c5e3521d6b96d46e0056d5d09bd411c025b02c56b5fae7
Manifests, environment (5 files)	
logs/v8/manifest_XVIP.json	8dcd372a5977065d7f30e9698917c9b371faf5caa111a84bd9d27a2e4c2714a6
logs/v8/manifest_XVIQ.json	c33ef1abe8d6f63a28885a7d41bc70d27455cf45eba06a94697270d5cd0ab337
logs/v8/manifest_XVIR.json	44f17586766b4c13c088080143a99864b397dbc1893a7b39422855e88876fa17
logs/v8/tree_hashes_v8.json	a6b3246b70cf27eed7eb7d2f7f475903fbb8d1cd7199283ed1b90b5f42bceb8f

File	SHA-256
logs/v8/env.json	4a40f99a724e190dad2fa6e65c49b318415204377fa0723aa71128884aed714e

C Source, version, and compilation declarations

Supplied material. The v6 manuscript and the Python-listing manuscript were supplied in full as text. Statements about their historical executions are attributed to those sources. In v8 the frozen XVI-P/XVI-Q engine files were read and replayed from read-only copies; nothing in those trees was edited (Section 13.6). v7 was read in full.

Companion source scope. The I-B and I-B-G public records support the host and test-domain comparisons cited above. The other Cheung companion descriptions are cited at the version and locator level supplied with the revision material. Their full manuscripts and proofs are not represented as independently audited here. In particular, the source-qualified tensor-core and XVI-O1-ST comparisons are not silently upgraded to local proofs.

Primary analytical sources. The Bessel representation, standard special-function identities, and self-adjoint functional calculus are identified inputs. The normalizations, subtractions, Mellin reduction, and finite catalogue algebra used in the conclusions are written out above.

Code status. v7: the embedded program is newly consolidated source. v8: it was extracted unchanged and executed, together with the v8 programs in `code/v8/`; return codes, logs and SHA-256 values are in Appendix B. Software: Python 3 with SymPy (versions in `logs/v8/env.json`). No proof-assistant check is asserted.

Typesetting. This is a single-file pdfLaTeX source with an internal bibliography and inline listings. It requires no external figure, bibliography database, code file, or shell escape for typesetting. v8 was compiled with

```
pdflatex -halt-on-error XVI-O1-CONE-v8.tex
pdflatex -halt-on-error XVI-O1-CONE-v8.tex
```

and the final log was checked for unresolved references and citations.

Authorship and approval. The author line follows the supplied manuscript. This revision does not independently authenticate authorship, archive provenance, or author approval. No independent review of v8 has taken place; nothing was pushed or posted. Final submission requires the author's approval of the claims, scope, versions, and computational artifacts.

Standing non-claims. The cone, deficit, holonomy, extension, and CR branch remain prescribed inputs. No Einstein evolution, physical tensor Hessian, spacetime emergence, gravitational evaporation channel, or inherited characteristic quantum coefficient is constructed. XVI-N is not discharged; XV-B is not opened; H2/I3/J2 are not completed by this paper.

References

Marks (v8): * read in full; ‡ confirmed online (registry fetch of 18 October 2026, https://enter.grok.me/browse); † from memory or from a supplied description only.

[1] * N. Cheung, *Scoped O.1 on the Prescribed Cone, Version 7*, 18 October 2026. Source `XVI-O1-CONE-v7.tex` (SHA-256 `d0ba7b9c...5d132`); the base of this revision.

[2] * N. Cheung, *Scoped O.1 on the Cone, v6: Executed Apex Records, a Closed Constant-Profile Slavnov–Taylor Row, Weighted Apex Coefficients, and Ward-Identity Boundaries*, 18 October 2026. Supplied complete source `XVI-O1-CONE-v6.tex`. Historical execution and manifest claims are attributed to this source, not newly authenticated here.

[3] * N. Cheung, *XVI-O1-CONE: Python scripts (v1–v6)*, 18 October 2026. Supplied complete listing source `XVI-O1-CONE-scripts.tex`. The listings state that frozen XVI-P/Q files and run logs are external dependencies.

[4] ‡N. Cheung, *XVI-P*, Enter Calculus Companion, unpublished manuscript and computational records. Frozen p22 contract as identified in the supplied cone audit; later p23 excerpt is a comparison, not a silent contract change. Named engine files: `code/p5_entry/amplitude_rules.py`, `feynman.py`, and `dred.py`. Relevant supplied locators include `rem:o1`, `rem:p5`, and `rem:qds`.

[5] ‡N. Cheung, *XVI-Q*, Enter Calculus Companion, unpublished records. Supplied Q02 graph-normalization description and separately scoped Q11 release. Named comparison artifacts include `Q02_bulk_graphs.json` and `P18_D11.json`. v8 inspected and replayed named engine files from a read-only copy.

[6] †N. Cheung, *XVI-R*, frozen comparison record identified in the supplied cone audit. No theorem from its unavailable contents is imported here.

[7] ‡N. Cheung, *I-B: A Finite Enter Spacetime: Calibrated Destructive Time Readout on a Prescribed Conical Product*, version 1, Enter Calculus Companion. Relevant locators: `thm:instrument`, `prop:not-source`, `prop:no-return`, and `prop:record-residual`. Public manuscript record: https://enter.grok.me/browse/I-B.

[8] ‡N. Cheung, *I-B-G: Intrinsic and Holonomy-Twisted Green Operators on a Prescribed Conical Spacetime*, version 2, Enter Calculus Companion. Relevant locators: `prop:nonGH`, `thm:green`, `thm:speed`, and `rem:compact-exact`. Public manuscript record: https://enter.grok.me/browse/I-B-G.

[9] ‡N. Cheung, *I-B-G-S: Intrinsic Spin-1 and Spin-2 Green Operators on the Prescribed Conical Spacetime*, version 2, Enter Calculus Companion. Supplied version-qualified excerpts, especially `thm:spin1`, `thm:spin2`, `def:geo`, and `thm:selection`. Free source-test compatibility is not an interacting loop theorem.

[10] ‡N. Cheung, *I-B-G-S-B*, Enter Calculus Companion. Supplied source descriptions of positive tensor-core selection and exceptional source-test defects. Relevant supplied locators: `cor:exclude`, `thm:contrast`, and `prop:warddefect`. A complete independent tensor-core proof is not asserted here.

[11] ‡N. Cheung, *I-B-VP: Vacuum Polarization of the Friedrichs Ground State on a Prescribed Conical Spacetime*, version 3 and supplied predecessor passages, Enter Calculus Companion. Relevant supplied locators: `lem:infrared` and `thm:friedrichs`. One-point/compact-test comparison only.

[12] ‡N. Cheung, *I-B-VP-1: Prescribed Apex Finite Parts on a Conical Enter Spacetime: Wick Square, Stress Tensor, and Charged Current*, version 3, October 2026. Supplied source-qualified descriptions of one-point extension freedom and Ward conditions.

[13] ‡N. Cheung, *I-B-VP-4*, Enter Calculus Companion. Supplied frozen-source Maxwell-response comparison. The current and its Maxwell response are distinct objects; no interacting cone Ward theorem is imported.

[14] ‡N. Cheung, *I-B-VP-5-Ex1-B*, version 1, 18 October 2026. Supplied excerpts on heat normalization, local stress running, continuity of complete finite parts, and non-universality. Relevant locators: `thm:running`, `thm:continuity`, `thm:equalradius`, and `thm:trace`.

[15] ‡N. Cheung, *I-B-VP-5-Ex2: Smooth Flux-Core Selection on a Prescribed Cone*, version 1, 18 October 2026. Supplied excerpts on twisted scalar Friedrichs selection and critical-window Dirac selection.

[16] ‡N. Cheung, *I-B-VP-5-Ex2-B*, version 1, 18 October 2026. Supplied excerpts on exhausted-annulus Mosco arguments, endpoint continuity, and finite-core qualifications.

[17] ‡N. Cheung, *I-B-ECHO-v2: Diffracted Echoes and Finite Supplied-Report Instruments*, version 2, October 2026. Supplied source-qualified comparisons: `thm:readout`, `thm:core`, and `cor:nonreconstruction`. No continuum acquisition or loop-residue identity is imported.

[18] ‡N. Cheung, *I-C-v2: The Shredder Interface: Destructive Readout, Black-Hole Analogues, and Quantum Computers*, version 2, October 2026. Supplied excerpts and source descriptions. Theorems A–F separate referenced recovery, classical histories, conditional thermal accounting, complete classical access, endpoint underdetermination, and persistent classical output.

[19] ‡N. Cheung, *En09-A: Realization of Relational Time by Destructive Enter on a Clock Register*, version 1, Enter Calculus Companion. Supplied excerpts, especially `thm:main`. Complete reference-preserving instrument and consumed-clock conditioning; no continuum acquisition is inferred.

[20] ‡N. Cheung, *En07-A*, *En07-B*, and *En07-C*, version 1 source records, Enter Calculus Companion. Supplied complete-complement and classical-output comparisons. Model-specific channels and physical endpoint assumptions are not imported.

[21] ‡N. Cheung, *X-J-ST-2*, Enter Calculus Companion. Identifier and `prop:noacquisition` cited from the supplied revision memorandum. No complete manuscript or additional theorem is asserted to have been inspected.

[22] ‡N. Cheung, *Oai290-A*, version 1, Enter Calculus Companion. Supplied excerpts distinguish whole-code recovery, named observable targets, and declared geometric models. The companion identifier does not make an external catalogue family a premise of this apex calculation.

[23] ‡N. Cheung, *XVI-O1-ST-v5*, Enter Calculus Companion. Source-qualified sign-integrity witness described in the supplied revision memorandum: D01 sixth coordinate $-4/3$ versus $+4/3$, with difference $(8/3)e_6$. Full underlying artifacts were not supplied for this revision.

[24] ‡*Hawk05-A*, *Hawk05-B*, and *Hawk05-C*, version 1 supplied manuscript records, attributed to N. Cheung in the supplied source list. Finite operational recovery, Page–Sen proxy, and conditional extrapolation comparisons only. Their authorship and full artifacts are not independently authenticated here; no gravitational Page theorem is imported.

[25] ‡N. Cheung, *XVI-N*, Enter Calculus Companion. Supplied version-qualified boundary, including the identifier `thm:massless-obstruction`. No absent theorem is reconstructed or discharged here.

[26] ‡N. Cheung, *XIV-H*, *XIV-I*, and *XIV-J*, version 1 source records, Enter Calculus Companion. Supplied historical H2 ending, narrower I2 construction, and unresolved full I3/J2 characteristic-map obligations.

[27] ‡N. Cheung, *XV-B*, version 2, Enter Calculus Companion. Supplied technical-interface descriptions. The technical manuscript is distinguished from the unopened downstream quantum-admission exit.

[28] †D. V. Fursaev, "The heat kernel expansion on a cone and quantum fields near cosmic strings," *Classical and Quantum Gravity* **11** (1994), 1431. DOI: 10.1088/0264-9381/11/6/008. arXiv:hep-th/9309050. The full-opening-angle convention and equations (2.11)–(2.15) fix the comparison in Remark 5.5.

[29] †D. V. Fursaev and G. Miele, "Cones, spins and heat kernels," *Nuclear Physics B* **484** (1997), 697–723. DOI: 10.1016/S0550-3213(96)00631-1. arXiv:hep-th/9605153.

[30] †D. V. Vassilevich, "Heat kernel expansion: user's manual," *Physics Reports* **388** (2003), 279–360. DOI: 10.1016/j.physrep.2003.09.002. arXiv:hep-th/0306138.

[31] †F. W. J. Olver et al., editors, *NIST Digital Library of Mathematical Functions*, Chapter 10, especially Sections 10.27, 10.29, 10.32, and 10.43. Modified-Bessel relations, recurrences, and integral transforms. https://dlmf.nist.gov/10.

[32] †J. Dereziński and S. Richard, "On Schrödinger operators with inverse square potentials on the half-line," *Annales Henri Poincaré* **18** (2017), 869–928. DOI: 10.1007/s00023-016-0520-7. arXiv:1604.03340.

[33] †B. S. Kay and U. M. Studer, "Boundary conditions for quantum mechanics on cones and fields around cosmic strings," *Communications in Mathematical Physics* **139** (1991), 103–139. DOI: 10.1007/BF02102731.

[34] †D. Stöckinger, "Regularization by dimensional reduction: Consistency, quantum action principle, and supersymmetry," *Journal of High Energy Physics* **2005**(03), 076. DOI: 10.1088/1126-6708/2005/03/076. arXiv:hep-ph/0503129.

[35] †M. Horodecki, P. W. Shor, and M. B. Ruskai, "Entanglement breaking channels," *Reviews in Mathematical Physics* **15** (2003), 629–641. DOI: 10.1142/S0129055X03001709. Preprint title: "General Entanglement Breaking Channels," arXiv:quant-ph/0302031.

[36] †M. A. Nielsen, "A simple formula for the average gate fidelity of a quantum dynamical operation," *Physics Letters A* **303** (2002), 249–252. DOI: 10.1016/S0375-9601(02)01272-0. arXiv:quant-ph/0205035.

[37] †D. Reeb and M. M. Wolf, "An improved Landauer principle with finite-size corrections," *New Journal of Physics* **16** (2014), 103011. DOI: 10.1088/1367-2630/16/10/103011. arXiv:1306.4352.